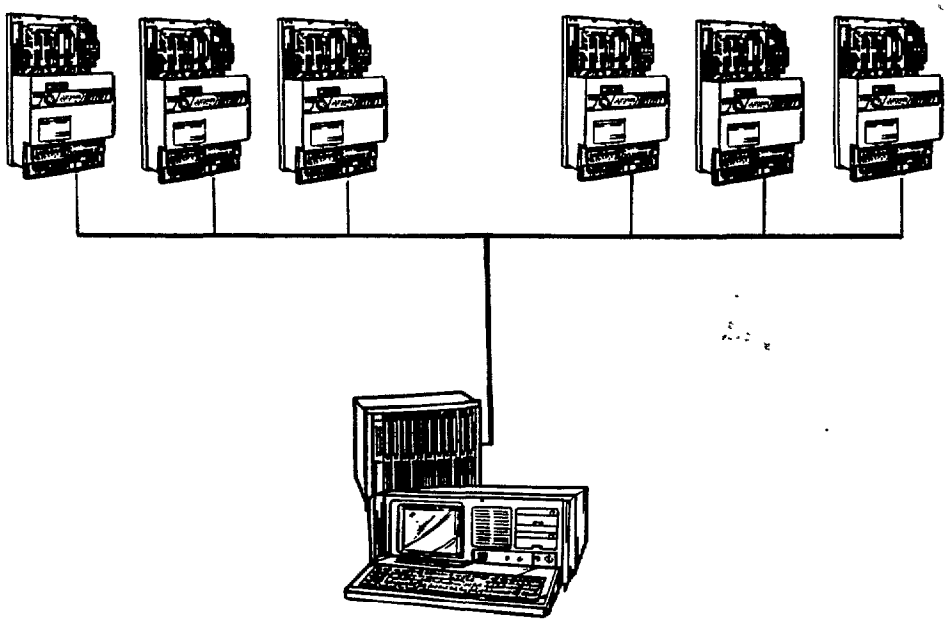
Preliminary GEK-85789

GE DRIVE SYSTEMS

INSTRUCTIONS



LOCAL AREA NETWORK

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GE Drive Systems

INSTRUCTIONS

IMPORTANT

As of **August 1992, GE Drives Products Operation** (located in Erie, Pennsylvania) has been relocated and integrated with **GE Drives Systems** (located in Salem, Virginia). Within this publication there are phone numbers for contacting the factory when **Product Service Engineering** assistance is needed.

The following phone numbers should be used after **August 1992** instead of the ones currently listed in this publication.

During Normal Working Hours (8 AM - 5 PM EST):	**(703) 387-7595**
After Hours for **Emergency Assistance Only:** **NOTE:** *This "After Hours" number will activate a Voice Menu. Make your desired selection as directed to be placed in contact with the proper person to assist you.*	**(703) 387-8080**

When seeking assistance, use the **Serial Number, Requisition Number, Model Number,** and **Drive Code** to identify the equipment. Please have this information available prior to placing a call.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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1. SAFETY PRECAUTIONS AND RECEIVING/STORAGE GUIDELINES

WARNING, CAUTION AND NOTE LABELS PLACED ON THE EQUIPMENT

The following format is used for the safety and informative labels placed on the equipment. Read all labels and follow the directions of them whenever working on the equipment.

WARNING: Denotes operating procedures and practices that may result in personal injury or loss of life if not correctly followed.
WARNING labels will be red in color with black or white lettering.

CAUTION: Denotes operating procedures and practices that, if not strictly observed, may result in damage to, or destruction of the equipment.
CAUTION labels will be amber in color with black lettering.

NOTE: Notes call attention to information that is especially significant in understanding and operating the equipment.
NOTE labels will be white in color with black lettering.

WARNING, CAUTION AND NOTE PARAGRAPHS WITHIN THIS INSTRUCTION

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment. These safety precautions will be repeated throughout this instruction book where applicable.

WARNING - SHOCK AND LACERATION HAZARD: Danger of injury from rotating equipment may exist due to remote energization or operation of the equipment.

WARNING - SHOCK AND LACERATION HAZARD: Assure safety of personnel throughout the entire system before operating the LAN and associated equipment.

WARNING - SHOCK AND LACERATION HAZARD: No modifications to the LAN equipment or software should be attempted without first reviewing the effect on the overall system and taking the necessary precautions to assure safety of personnel.

CAUTION: Do not remove connections or printed circuit cards from the controller while power is applied. This can damage the equipment.

CAUTION: Read and heed all WARNING, CAUTION, and NOTE labels posted on the qquipment or listed in the instructions and drawings.

NOTE: *The Local Area Network, the Series Six Programmable Controller, the Bus Interface Unit, and all the associated software form part of an overall drive system, including motor drive controllers, motors and rotating equipment.*

NOTE: *Always read the complete instructions prior to applying power or troubleshooting the equipment. Follow all procedures step-by-step.*

RECEIVING

The equipment and software should be placed under adequate cover immediately upon receipt as packing cases are not suitable for outdoor or unprotected storage. Each shipment should be carefully examined upon arrival and checked against the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from:

GE DRIVE SYSTEMS
ERIE, PA
(814) 875-3851

NOTE: *When requesting assistance, please refer to the serial number and model number to identify the equipment.*

STORAGE

If not to be installed immediately, hardware should be stored in a clean, dry location at ambient temperatures from -20 C to 55 C (-4 F to 131 F). The surrounding air must be free of chemically corrosive and electrically conductive contaminants. Care should be taken to protect sensitive solid state equipment from static electricity. Precautions should be taken to prevent condensation from forming within the equipment enclosure.

Software provided on magnetic media, such as floppy disks and magnetic tape, should be stored in a clean, dry location at ambient temperatures of from 10 C to 55 C (50 F to 125 F). Care should be taken to protect the media from magnetic fields, high heat or humidity and physical damage.

2. GENERAL DESCRIPTION AND COMPONENT FUNCTIONS

The GE DRIVE SYSTEMS Local Area Network (LAN) is an integral part of a coordinated drive system. These instructions provide information useful in installing, operating and maintaining the LAN. Further information is provided in additional instruction books specific to other system equipment and in the drawings and Custom Instructions provided with the equipment. The operators and persons responsible for maintaining the equipment should have access to a copy of these instructions at all times and should consult these instructions prior to operation or maintenance of the equipment.

The Local Area Network (LAN) provides serial data communications between the Series Six™ Programmable Logic Controller (PLC) and GE DRIVE SYSTEMS products such as the DC-300™, AC-300™ and AC-500™ Adjustable Speed Drives. Unlike point-to-point serial links, with dedicated transmit and receive connections for each device, the LAN allows multiple drives to be addressed over a single device (drop), sent to a select number of drops (zone or group) or broadcast to all drops on the LAN channel. The LAN may be used for automated control and monitoring of drive operation, including data logging, recipe down-loading, diagnostics, and history reporting.

The LAN utilizes a Bus Interface Unit (BIU) card to interface between the SERIES SIX Central Processor Unit (CPU) and the external members on the network. "One-Shot" commands issued through applications software within the PLC are communicated via DPREQ window requests to the BIU. These commands are interpreted by the BIU for implementation and communicated to the motor drives on the network. A Microapplication card in the drive controller interfaces the drive to the network. Essentially, any GE DRIVE SYSTEMS product which accepts the Microapplication card may be addressed over the LAN.

BUS INTERFACE UNIT (BIU Card, 531X175SSBA_M_)

The Bus Interface Unit (BIU) provides the necessary interface for communications between the Series Six PLC and the GE DRIVE SYSTEMS Local Area Network. Two serial communications channels, designated as channels P and Q, are supported by each BIU card. Up to 32 drops, including the BIU, may be addressed per channel.

A simplified representation of the BIU architecture is shown in FIGURE 2. The BIU card includes two programmable communications interfaces, each with its own dedicated microprocessor (SUP-P and SUP-Q), memory, transmitter, and receiver. The Series Six bus interface and BIU reset circuitry are shared common to both BIU channels.

Configuration Jumpers: Reconnectable jumpers are provided on the BIU card for configuration and test. Most jumpers will be factory set in the 1-2 position after test and should not be changed. Several jumpers select specific operating characteristics as described in the following paragraphs:

Jumpers JP19 and JP21 are preset based on BIU on-board memory size for channels P and Q respectively, and should also not be changed.

™ - *Series Six, DC-300, AC-300 and AC-500 are trademarks of General Electric Company, USA.*

TABLE 1: BIU - SERIES SIX PROGRAMMABLE CONTROLLER BACKPLANE DIP SWITCH SETTINGS

DPREQ Register Content		Switch Position
CHNL P	CHNL Q	7654321
1001	1009	------x
1017	1025	-----xx
1033	1041	----x-x
1049	1057	----xxx
1065	1073	---x--x
1081	1089	---x-xx
1097	1105	---xx-x
1113	1121	---xxxx
1129	1137	--x---x
1145	1153	--x--xx
1161	1169	--x-x-x
1177	1185	--x-xxx
1193	1201	--xx--x
1209	1217	--xx-xx
1225	1233	--xxx-x
1241	1249	--xxxxx
1257	1265	-x----x
1273	1281	-x---xx
1289	1297	-x--x-x
1305	1313	-x--xxx
x1321	1329	-x-x--x
1337	1345	-x-x-xx
1353	1361	-x-xx-x
1369	1377	-x-xxxx
1385	1393	-xx---x
1401	1409	-xx--xx
1417	1425	-xx-x-x
1433	1441	-xx-xxx
1449	1457	-xxx--x
1465	1473	-xxx-xx
1481	1489	-xxxx-x

DPREQ Register Content		Switch Position
CHNL P	CHNL Q	7654321
1497	1505	-xxxxxx
1513	1521	x-----x
1529	1537	x----xx
1545	1553	x---x-x
1561	1569	x---xxx
1577	1585	x--x--x
1593	1601	x--x-xx
1609	1617	x--xx-x
1625	1633	x--xxxx
1641	1649	x-x---x
1657	1665	x-x--xx
1673	1681	x-x-x-x
1689	1697	x-x-xxx
1705	1713	x-xx--x
1721	1729	x-xx-xx
1737	1745	x-xxx-x
1753	1761	x-xxxxx
1769	1777	xx----x
1785	1793	xx---xx
1801	1809	xx--x-
1817	1825	xx--xxx
1833	1841	xx-x--x
1849	1857	xx-x-xx
1865	1873	xx-xx-x
1881	1889	xx-xxxx
1897	1905	xxx---x
1913	1921	xxx--xx
1929	1937	xxx-x-x
1945	1953	xxx-xxx
1961	1969	xxxx--x
1977	1985	xxxx-xx

NOTES:

1. *DPREQ Register contents are expressed in decimal notation.*
2. *x = Switch in OPEN position (depressed to the left).*
3. *Switch #1 should be in the OPEN position for all conditions.*
4. *Selection of a P & Q channel address actually overlays the selected physical address plus the next 7 locations. (i.e., switch setting -----xx assigns I/O addresses 17 through 24 to P channel and 25 through 32 to Q channel). Care should be taken to not assign other physical uses of these addresses which would cause Series Six Programmable Logic Controller backplane contention problems.*

Jumpers JP3, JP4, JP14 and JP15 select network terminating resistors which must be used if the BIU is either the first or last connected device on the channel. Both jumpers associated with a given channel (JP3 & JP4 for Channel P or JP14 & JP15 for Channel Q) must be set in the same position (1-2) to insert the terminating resistor for that channel. Jumper position 2-3 disconnects the terminating resistors from the channel.

Jumpers JP47 through JP52 are used to select between standard LAN (1-2 position) or Isolated LAN (2 - 3 position). Refer to Isolated LAN paragraph of this instruction book for additional information.

BIU Address: The LAN protocol requires that each drop on the network be identified by a unique address. Default configurations for the SUP processors define the BIU address for each SUP as Drop 0 in Group 0 on their respective channels. Alternate configurations may be programmed through software in the Series Six PLC and must be down-loaded following each power up sequence or after a BIU hard reset. For more information on BIU addressing, refer to the Communications Protocols paragraph of this instruction book.

Location and I/O Addressing: The BIU card installs in any addressable I/O slot of a Series Six High-Capacity I/O Rack or Model 60 CPU Rack. Addressing for the BIU is configured by setting the DIP switches on the rack backplane as indicated in TABLE 1.

MICROAPPLICATION CARD (531X139APMA_M_)

The Microapplication card provides the necessary interface for communications between GE DRIVE SYSTEMS Local Area Network and the drive controller. This programmable card is available as an option on GE DRIVE SYSTEMS microprocessor based controllers and may be used to provide interfacing and programming for a number of specific applications, including LAN communications.

The Microapplication card contains a variety of analog and digital circuitry including a microprocessor (MUP), memory, transmitter and receiver. A shared bus allows MUP to communicate with the drives's regulator microprocessor (RUP).

Configuration Jumpers: Reconnectable jumpers are provided on the Microapplication card for configuration and test. The Microapplication cards on drives supplied as part of a LAN based system should already be configured at the factory. Most jumpers will be factory set after test and should not be changed. Several jumpers select specific operating characteristics, as described in the following paragraphs:

Jumpers JP18 and JP19 select network terminating resistors which must be used if the drive is either the first or last connected device on the channel circuit. Both jumpers must be set in the same position (1-2) to insert the terminating resistor. Jumper position 2-3 disconnects the terminating resistor from the circuit.

Jumpers JP58 through JP60 are used to select between standard LAN (1-2 position) or Isolated LAN (2-3 POSITION) operation. Refer to the Isolated LAN paragraph of this instruction book for additional information.

Other Microapplication card jumpers associated with the LAN function include **JP13, JP15, JP16, and JP17** which must be in the 1-2 position for proper operation. Generally, jumpers **JP38 and JP39** are placed in the 1-2 position to select communications using LANTB.

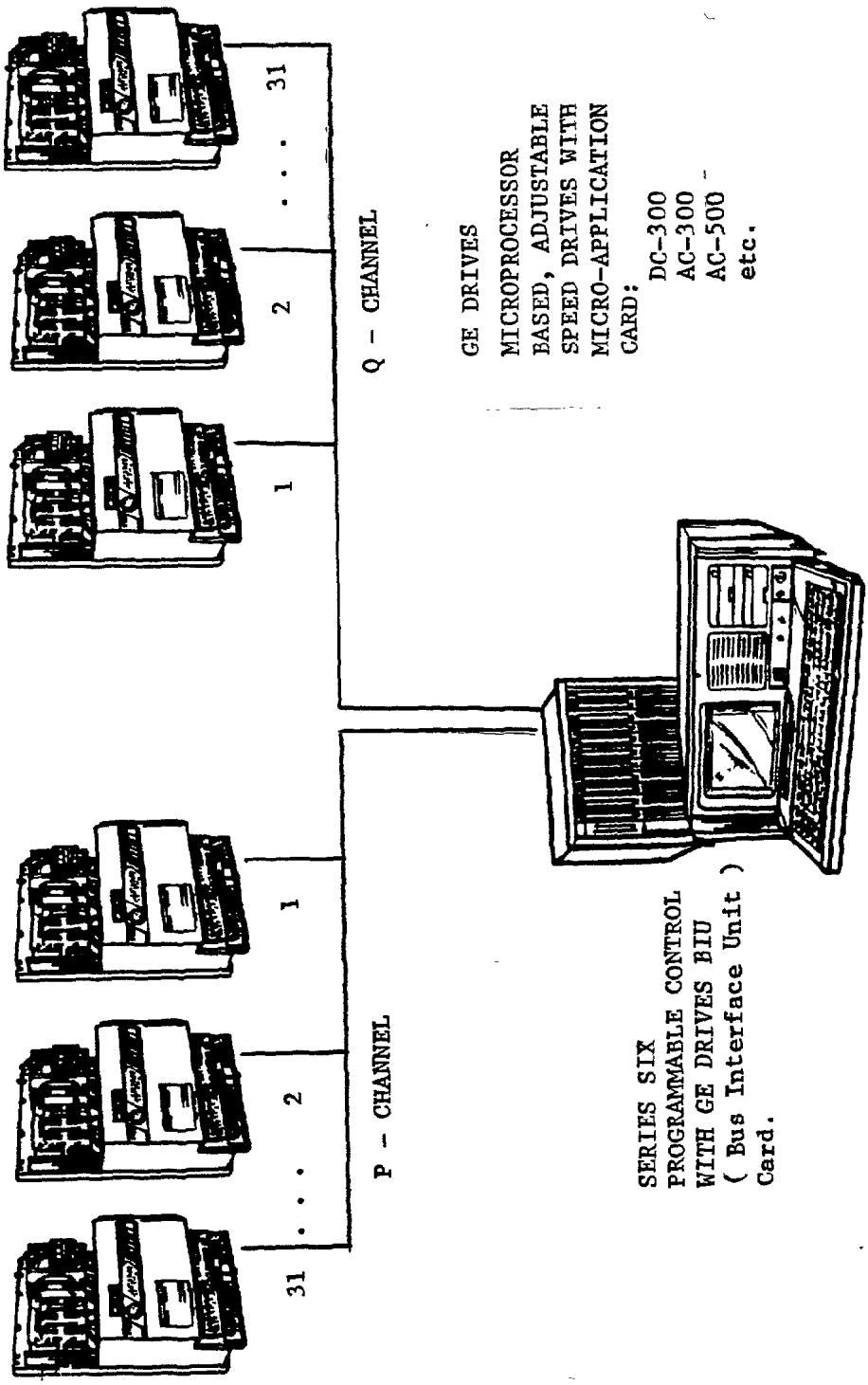


FIGURE 1. Local Area Network

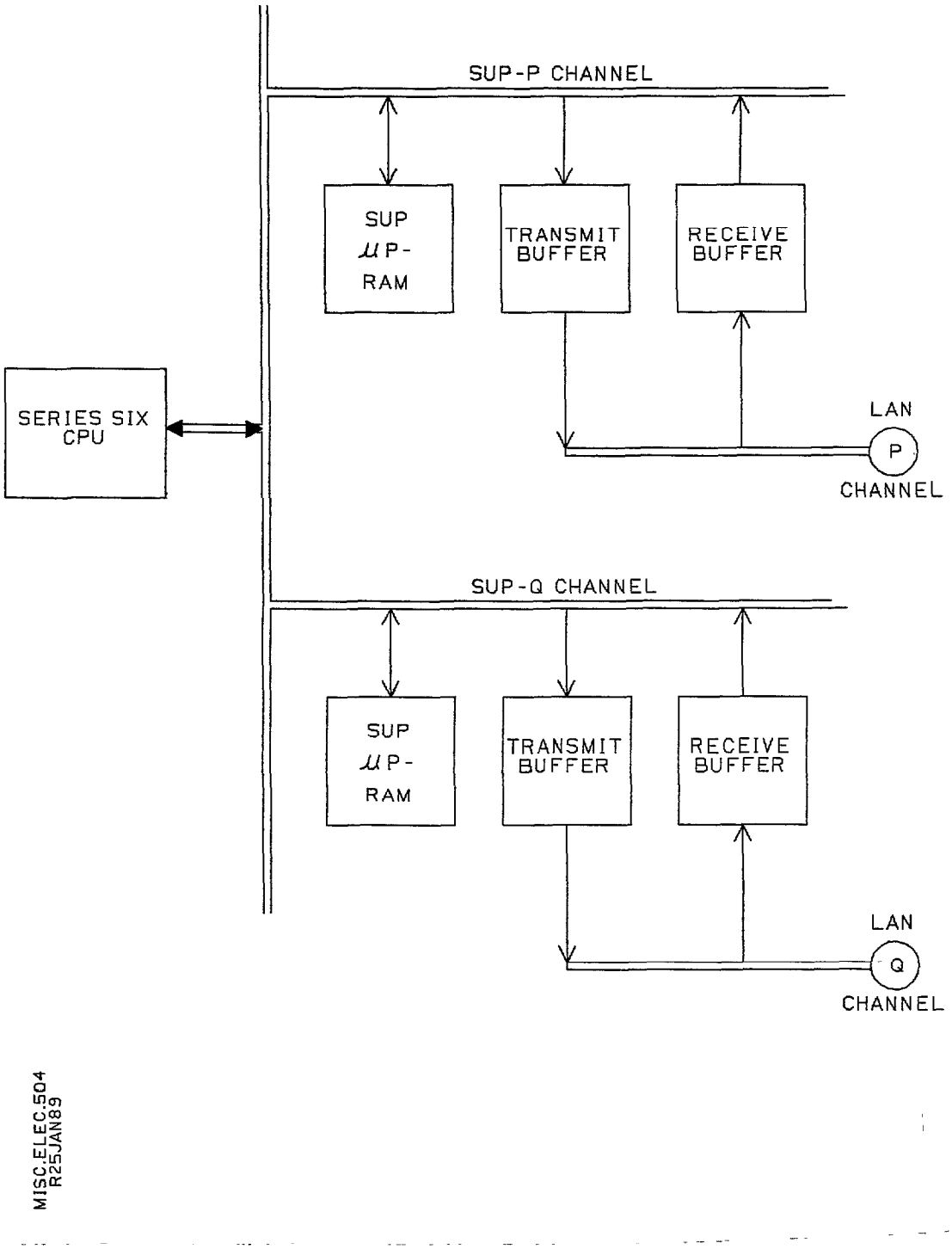


FIGURE 2. BIU Architecture

TABLE 2 provides a partial listing of Microapplication card jumpers related to the LAN and a brief description of their function. A complete summary of the specific jumper settings used on each drive may be found in the Custom Instruction Book provided with the drive. Jumper selections should be verified against this list if the Microapplication card is replaced.

BIU Address and MUP Parameters: The LAN protocol requires that each drop on the network be identified by a unique address. The BIU address (Drop and Group) for the drive controller is set through a parameter (ADR213-MSET3) stored in the programmable memory (EEROM) of the drive controller. A partial list of additional parameters used to program other LAN options is given in TABLE 3. These are provided for reference only; additional bits in the data word may be set for options not shown. Actual values for these and other parameters used in conjunction with specific options are listed in the Custom Instruction Book provided with the drive. For more information on drive parameters and programming refer to the Custom Instruction Book provided with the drive.

Location and Connection: The Microapplication card installs in the drive controller card carrier. Interconnections between the Control card and the Microapplication card are made through ribbon cable 3PL. Connectors 11PL, 14PL, and 16PL provide input and output connections to the Microapplication card. Depending on selection of jumpers JP22, JP23, JP38, and JP39, the Microapplication card serial interface may be accessed through either 11PL or LANTB. Generally LANTB is selected and connected by twisted-pair wire to LNTB on the LAN balun assembly of the drive. Refer to the Instruction Book and Elementary Diagrams provided with the drive for additional information on Microapplication card installation and connections.

ISOLATED LAN

An Optically Isolated LAN option is available for applications where greater noise immunity and rejection of common mode voltages may be required, such as installations requiring LAN wiring runs in excess of 200 feet or installations using Drop-In Operator Stations.

The following **minimum** components levels are required to utilize the Isolated LAN option:

Microapplication card	531X139APMAAM7 or later revision
Bus Interface Unit	531X175SSBAJM2 or later revision
LAN Power Supply	36A359133AAG01 (one per channel)

LAN Power Supply: The LAN Power Supply provides a current source driver for the Isolated LAN. It is mounted in the same enclosure as the I/O Rack containing the BIU card it serves. Jumper JP1 selects for LAN operation (2-3 position). Potentiometer P1 is provided for trimming (+5 %) the power supply's voltage regulator. Input and output connections are as follows:

TERMINALS	FUNCTION
1-3	115 VAC Control Power
7-8	Voltage output (not used on Isolated LAN)
11-12	Isolated LAN TXA (terminal 11) and TXB (terminal 12)

Isolated LAN Jumper Selection: Isolated LAN operation is selected by setting the following jumpers in the 2-3 position:

Microapplication card jumpers	JP58, JP59, and JP60
Bus Interface Unit jumpers	JP47, JP48, JP49, JP50, JP51 and JP52
LAN Power Supply jumper	JP1

NOTE: *Other jumpers, their functions and selections are described in the Bus Interface Unit and Microapplication card paragraphs of this instruction book.*

LAN Wiring: Generally the LAN wiring and installation should be in accordance with the procedures and recommendations outlined in Section 3 of this instruction book. Use of the Isolated LAN does, however, extend the maximum distance from the current source (LAN Power Supply) to the drops. The maximum total capacitance should be held to less than 0.14 Microfarad.

MAXIMUM DISTANCE	MINIMUM WIRE SIZE (Twisted pair)
700 Feet	18 Gauge (AWG)
1100 Feet	16 Gauge (AWG)
1700 Feet	14 Gauge (AWG)
2800 Feet	12 Gauge (AWG)

TABLE 2: MICROAPPLICATION CARD LAN JUMPERS

Jumper	Description/Position
JP13	MUP Interrupt 0 Source Select - Allows selection of source of interrupt INTO for MUP. **1-2** For LAN serial communication. **2-3** For MUP encoder marker pulse detection
JP15	TX Driver Mode - Controls mode of serial driver. **1-2** Passive pull-up for multidrop LAN communications. **2-3** Passive or active pull-up as controlled by MUP (not used at present)
JP1	Serial Mode Select #1 - Partially controls serial link modes. **1-2** Connects TX to RX for half duplex differential operation at LANTB; used for LAN communications. **2-3** Connects RX to JP19-3 for full duplex single-ended operation at LANTB, or for inhibiting serial communication, or for use as a non-serial input.
JP17	Serial Mode Select #2 - Partially controls serial link modes. **1-2** Connects /TX to /RX for half duplex differential LAN operation at LANTB. **2-3** Connects /RX to common to inhibit serial communication. **OPEN** Allows full duplex single-ended or non-serial operation at LANTB.
JP18	Serial Mode Select #3 - Partially controls Serial link termination **1-2** Connects terminating resistor (for first and last drops of differential LAN system) **2-3** Disconnects terminating resistor.
JP19	Serial Mode Select #4 - Partially controls serial link termination. **1-2** Connects terminating resistor (for first and last drops of differential LAN system). **2-3** Disconnects terminating resistor. Connects JP16-3 to 1.8 volt bias for single-ended operation.
JP22	11PL-1 Select - Allows multiple use of 11PL-1 input. **1-2** Encoder marker or external interrupt non-invertig input. **2-3** TX (alternate LANTB-1)
JP23	11PL-2 Select - Allows multiple use of 11PL-2 input. **1-2** Encoder marker inverting input. **2-3** /RX (alternate LANTB-2).
JP38	LAN source select - Allows LAN interface through 11PL instead of through LANTB. **1-2** Normal operation through LANTB. **2-3** Operation through 11PL-1 (also check JP22)
J39	LAN source select - Allows LAN interface through 11PL instead of through LANTB. **1-2** Normal operation through LANTB. **2-3** Operation through 11PL-2 (also check JP23).

TABLE 2: MICROAPPLICATION CARD LAN JUMPERS (continued)

Jumper	Description/Position
JP42	MUP timer/external interrupt mode selection - Allows various application options with regard to this MUP resource. **1-2** MUP encoder operation without a LAN (U30=8032) or with a LAN (U30-80C252). **2-3** External interrupt from EBR when with a LAN and without the MUP encoder. **2-4** For external pulse counting capability from EBA with or without the LAN when the MUP encoder is not used.
JP58 JP59 JP60	These jumpers provide LAN operation mode-selects for standard or isolated LAN connection. Refer to the Isolated LAN paragraph of this instruction book for details. **1-2** Standard LAN operation. **2-3** Isolated LAN operation.

TABLE 3: MICRO APPLICATION CARD LAN SET-UP PARAMETERS

Name	Adr.	Description		
		Select Bit Settings	Dec. Wt.	Function
APPJP	010	**Application Card selection jumpers:**		
		Bit 0 indicates if an Application or Microapplication Card is installed in the drive. Should be set (1).		
		0	000	Not Installed
		1	001	Card is installed
SPCJP	014	**Special application jumpers:** Bit 7 enables MUP-RUP interface between the Microapplication Card and the drives regulator microprocessor (RUP). Should be set (1).		
		.... 0...	000	Disable MUP-RUP interface
		.... 1...	128	Enable MUP-RUP interface
MSET1	211	**Microapplication Card Set-Up Parameter #1:**		
		Bits 0 - 4 select drive **Model Identification** to be communicated to the LAN.		
		....0 0001	001	DC-300 drive
		....0 0011	003	AC-300 drive
		....0 0100	004	AC-500 drive
		Bit 8 enables the **LAN interface.** Normally cleared (0).		
		....0	000	Enable LAN
		....1	256	Disable LAN
		Bit 9 selects protocol for **Weighted Zero** data bytes and expected **Acknowledgments** from the drop on group or broadcast messages.		
		.... ..0.	000	Zero/Acknowledgement not used
		.... ..1.	512	Zero/Acknowledgement selected
MSET2	212	**Microapplication Card Set-Up Parameter #2:**		
		Bits 8 - 15 select LAN communications **Baud Rate.** Normally set for 57.6 baud.		
		0100 0000	16384	19200 Baud Rate
		1000 0000	32768	57600 Baud Rate

TABLE 3: MICRO APPLICATION CARD LAN SET-UP PARAMETERS (continued)

Name	Adr.	Description		
		Select Bit Settings	Dec. Wt.	Function
MSET3	213	**Microapplication Card Set-Up Parameter #3:**		
		Bits 0 - 7 select drive's **Drop Number** to be used for identification and timing in LAN communications. Each device on a LAN channel must have a unique Drop Number from 0 to 31.		
		.... 000x xxxx	0 - 31	Drop Number
		Bits 8 - 15 select drive's **Group Number** to be used for identification in Group Addressed LAN communications. Each device to be addressed within a group on a LAN channel must be assigned to that Group Number (0 - 31). The decimal weight of the number entered is the product of the Group Number multiplied by the constant 256.		
		000x xxxx	0-7936	Group Number times 256
MSET4	214	**Microapplication Card Set-Up Parameter #4:**		
		Bits 0 - 5 select the **Channel Size**, indicating the total number of drops on the LAN channel, including the BIU.		
		Example: If 23 drives are active on channel P, the link size of 24 should be entered in all drives connected to channel P.		
		....xx xxxx	1 - 32	Number of drops on channel
MSET5	215	**Microapplication Card Set-Up Parameter #5:**		
		Selects up to six drive parameters which may be monitored, with the Fault Code, in a circular list that is accessed by the LAN FREEZE command. Refer to the Custom Instruction Book for the drive for options provided. The order of display is from the lowest bit weight to highest, with the Fault Code last.		
		1	001	Current
		....1.	002	Voltage
		.... .1..	004	Speed
		.... 1...	008	Field Current
		....1	016	Speed Reference
		....1.	032	RUP Status
		xxxx xxxx xx..	YYY	Custom Defined Parameters
UAPJP	251	**Microapplication Jumper:**		
		Bit 10 selects for **MUP Software** revision levels relative to REV 5.2.		
		.... .0..	0000	MUP earlier than REV 5.2
		.... .1..	01024	MUP REV 5.2

3. INSTALLATION/WIRING GUIDELINES AND COMMUNICATIONS

BIU CARD INSTALLATION

The BIU card may be installed in any I/O slot in a local Series Six High Capacity I/O Rack or in an I/O slot in a Model 60 CPU rack. A local I/O rack is defined as the CPU and I/O racks connected by a parallel cable with the most distant I/O rack no further than 2000 feet from the CPU. Backplane DIP switch settings are made according to the DPREQ address for the BIU as indicated in TABLE 1.

CAUTION: Relatively small amounts of excess charge can cause very intense electrostatic fields in metal - oxide semiconductor (MOS) devices, damaging their gating structure. Avoid handling circuit boards under conditions favoring the build-up of static electricity. Failure to observe this caution could result in the destruction of CMOS devices used in this equipment.

CAUTION: Backplane contention problems will occur if two or more cards are addressed at the same location as selected by the backplane DIP switches. Equipment misoperation or damage may result.

CAUTION: Do not remove connecitons or printed circuit cards from the equipment while power is applied as this may damage the equipment.

NOTE: *CPU software version V103 or later is required to implement the checksum feature of the window communications.*

CABLE SELECTION

A high quality cable, preferably one designed specifically for computer data communications, is recommended for wiring the Local Area Network. The LAN data rate of 57.6K BAUD and the expected length of cable runs should be considered when specifying the interface cable. Minimum specifications should include the following:

--- **Construction** - The cable should be twinaxial with twisted pair conductors and an overall shield. Insulation and temperature rating should be as appropriate for the industrial enviroment in which the cable is applied and to be in accordance with local codes.

--- **Capacitance** - A low capacitance cable should be used (typically between 8 pf/ft to 35 pf/ft) such that overall shunt capacitance is maintained below 0.032 microfarad.

--- **Wire Gauge** - Wire should be of sufficient gauge to provide the necessary mechanical strength as well as low overall voltage drop. Typically, 24 AWG wire is recommended as a minimum, with heavier gauges required for a longer runs or where mechanical stress is expected.

--- **Shielding** - The cable should incorporate a continuous overall shield with 100 % coverage. Typical computer communications cables which might be used for LAN wiring include Belden 9132 or Alpha 9823, 150 ohm twinaxial cable. For short runs, almost any 18 gauge twisted pair cable (run in its own conduit or isolated in free air) may suffice.

WIRING THE NETWORK

Prepare and connect the required interface cables between the Bus Interface Unit and drops (i.e. drives) on each channel. Refer to the instructions and figures contained in this publication, instruction books for related equipment, and the elementary diagrams provided for the system.

CAUTION: There are two seperate LAN channels for each BIU card. Care must be exercised to identify and differentiate between wiring for these two channels and to assure that the proper equipment is connected to the appropriate channel. Failure to do so will result in communication failures and system misoperation.

Wiring practices suitable for data communications and in accordance with local codes should be used; including the following considerations:

--- The Local Area Network members may be connected together in a "daisy chain", a "star" configuration or a combination of both as shown in FIGURE 3.

--- Cable runs should be kept as short as possible while adhering to proper routing practices for data communications wiring. Cable runs totalling over 200 Feet are not recommended without the use of the Isolated LAN option or the Serial Communications Isolator.

--- Data communications cables should be run in conduit or wireways seperate from other signal, control, or power wiring.

--- The cable shield should be directly connected to chassis (system) ground at one end only. To drain induced high frequency electrical "noise" while preventing circulating "ground loop" currents, the shield may be connected through capacitors to ground at other cable termination points.

CAUTION: The Series Six PLC does not isolate the LAN signal wires from its common. Therefore, the system common should be grounded and at the same potential as the PLC ground. Steps should be taken to assure that the BIU card and the LAN circuits to which it is connected are grounded to a common point. Failure to do so could result in damage to the equipment or misoperation.

CONNECTIONS

Drive controllers supplied by GE DRIVE SYSTEMS for LAN applications should have the Microapplication card installed at the factory. Generally internal LAN connections (shown in FIGURE 4) are prewired from LANTB on the card, through the ferrite core, and terminated at LNTB on the balun assembly that is attached to the drive controller. Field wiring should consist primarily of wiring between the BIU card and LNTB of the drives on the Local Area Network.

1. Connect BIU card - Connections for each channel to the terminal board on the BIU card front-panel connector block as follows:

   P Channel: White wire to terminal point 1 (TPA)
   Black wire to terminal point 2 (TPB)
   Shield to terminal point 3 (GND)

   Q Channel: White wire to terminal point 25 (TQA)
   Black wire to terminal point 26 (TQB)
   Shield to terminal point 27 (GND)

   Terminal point 6 (GND) should be tied to system ground (CCOM), as indicated in the system elementary diagrams.

2. Connect drive controller connections to the appropriate BIU channel for each drive at the terminal board on the LNTB Balun assembly attached to the drive controller:

   Drive LNTB: White wire to terminal point RX
   Black wire to terminal point TX
   Shield to terminal point S

   NOTE: *The shield should not be directly tied to chassis ground at the drive. Instead, a capacitor coupled ground is provided through LNTB terminal S to prevent ground loop currents from circulating through the shield.*

3. Connect terminating resistors to the last drop at the end of a run, as shown by the "T" designation in FIGURE 3. These resistors can be connected at the drive by placing Microapplication card jumpers JP18 and JP19 in the 1-2 position. Terminating resistors are connected at the BIU card by placing BIU card jumpers JP3 and JP4 (for channel P) or JP14 and JP15 (for channel Q) in the 1-2 position. To prevent excessive loading of the LAN, termination resistors should be used only where required with generally no more than five per channel.

NOTE: *In certain applications, the optically isolated LAN option may be used to permit longer wiring runs and provide improved noise immunity. Refer to the Isolated LAN paragraph of this instruction book for proper connections and associated jumper selections.*

COMMUNICATIONS PROTOCOLS

Two levels of communications connect the Series Six PLC, through the Bus Interface Unit, with the members on the Local Area Network:

LAN-BIU: Communications on the network between the drops and the Bus Interface Unit.
CPU-BIU: Communications between the Series Six CPU and the Bus Interface Unit.

Generally, LAN operations are transparent to the user and require no detailed knowledge of the communications protocols involved. The BIU card, Microapplication card, and LAN Standard software provided with the system handle all communications tasks. However, a general knowledge of the LAN protocol may prove useful in understanding overall system performance and in diagnosing system problems should they occur.

LAN-BIU COMMUNICATIONS: Each BIU channel may connect up to 32 drops together on the network with each member identified by a unique address (referred to as the BIU ADDRESS). The address includes a specific DROP NUMBER (0-31) for the individual drop and a GROUP NUMBER (0-31) which identifies the drop as a member of a preselected group of drops. Each drop on a given channel must have a unique Drop Number and can only be assigned to one group.

The BIU implements a simple "token passing" protocol in which each member on the channel is given a turn to transmit in order of address. If there is no drop at a particular address, it is automatically and quickly skipped over. Messages transmitted on the channel are addressed to the members in one of four ways:

DIRECTED: to a single specific drop identified by the Drop Number.
ZONE: to consecutive drops defined in Series Six software
GROUP: to a predetermined set of drops identified by the Group Number.
BROADCAST: to all drops on a given channel.

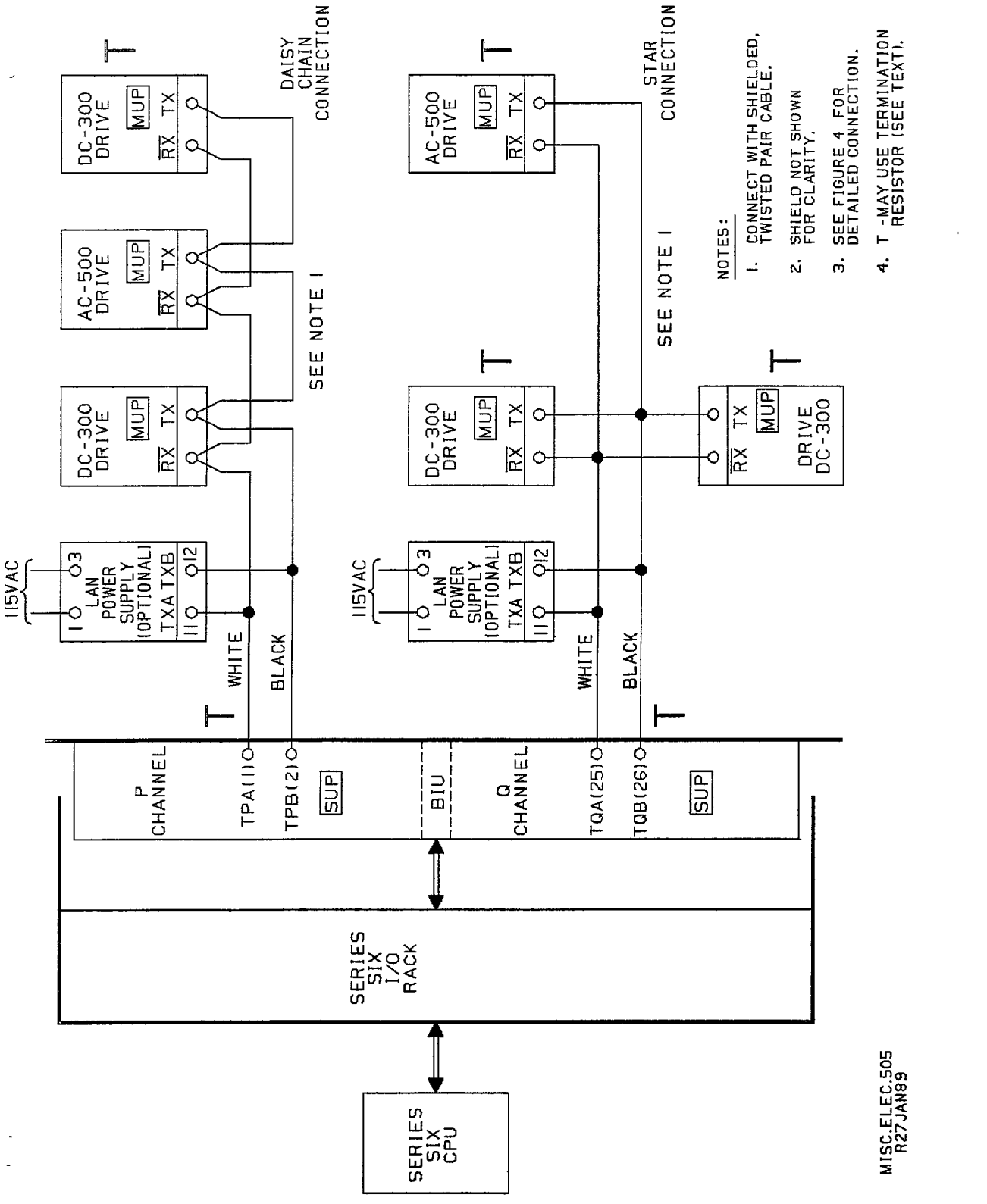


FIGURE 3. Typical LAN Configurations

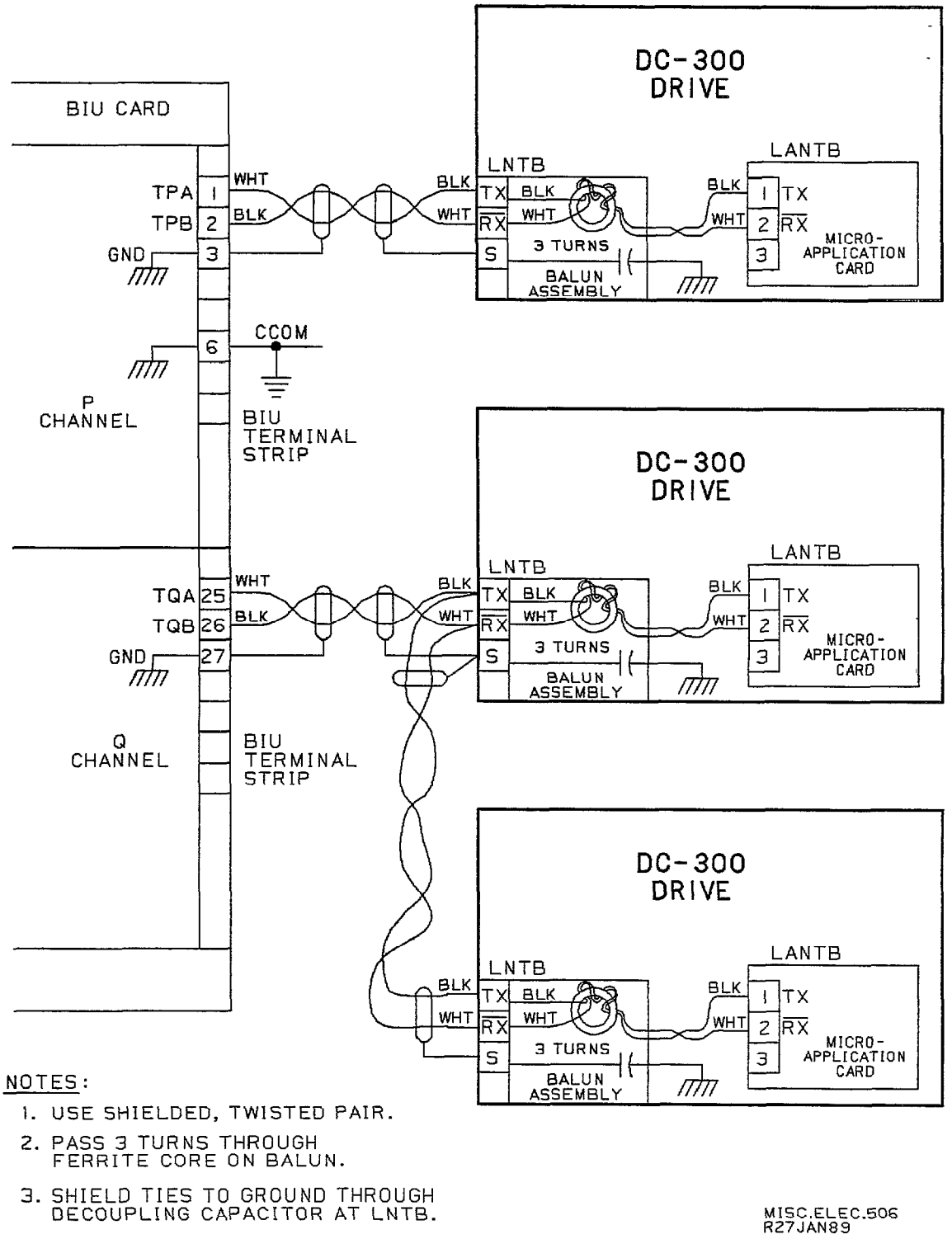


FIGURE 4. Typical LAN Connections

The message provides fields which may include Source (SRC), Acknowledgment (ACK), and Destination (DEST) addresses, Command (CMD), related drive memory Addresses (ADDR) and Data (DATA), Checksum (CKSM) and End Of Transmission (EOT) definitions. The number of fields and definitions used are determined as appropriate for the particular command and addressing being used. (FIGURE 5 shows typical format for messages on the LAN.)

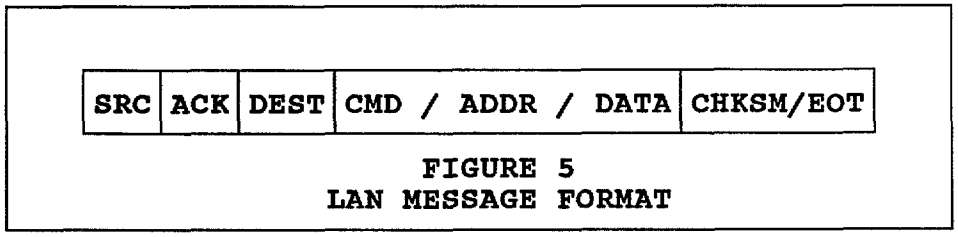


FIGURE 5
LAN MESSAGE FORMAT

Message integrity is assured through checksum validations and message format checking on a per-message basis. When a directed message is received and accepted, a message acknowledgment is sent to the source of the message. Acknowledgments are optional for zone, group, and broadcast messages. If no acknowledgment is echoed when expected, the message is retransmitted up to five times. Should transmission be unsuccessful, an error flag is set.

CPU-BIU COMMUNICATIONS: Communications between the BIU and the Series Six CPU are implemented through ladder logic in the programmable logic controller. The following discussion assumes some familiarity with Series Six ladder logic programming; experience with the ASCII/BASIC module may also prove helpful, as the window communications schemes that they use are similar.

Communications between the CPU and BIU may be initiated through data processing request (DPREQ) instructions. Following the execution of the instruction, a communications "window" is opened to allow data transfer between the CPU and the processor (SUP) of the selected BIU channel. At the end of every window communications cycle, data words indicating the LAN status and communications errors, if any, are returned by the Bus Interface Unit.

COMMAND BLOCK: Before executing the DPREQ instruction, a message Command Block is set up for the data transfer operation. The Command Block (FIGURE 6) is comprised of a command field, followed by data fields if required. The command field consists of a 16 bit BIU address followed by status words, error words, and two 8 bit parameters. The BIU address contains the I/O addresss for the BIU plus 1000 decimal. The most significant portion of this address matches the I/O address for the BIU card selected by the DIP switches on the I/O rack backplane. The least significant portion of the address indicates which BIU channel (P or Q) is to perform the communication. The 8 bit parameters are the SUP Command and the number of words contained in the data field.

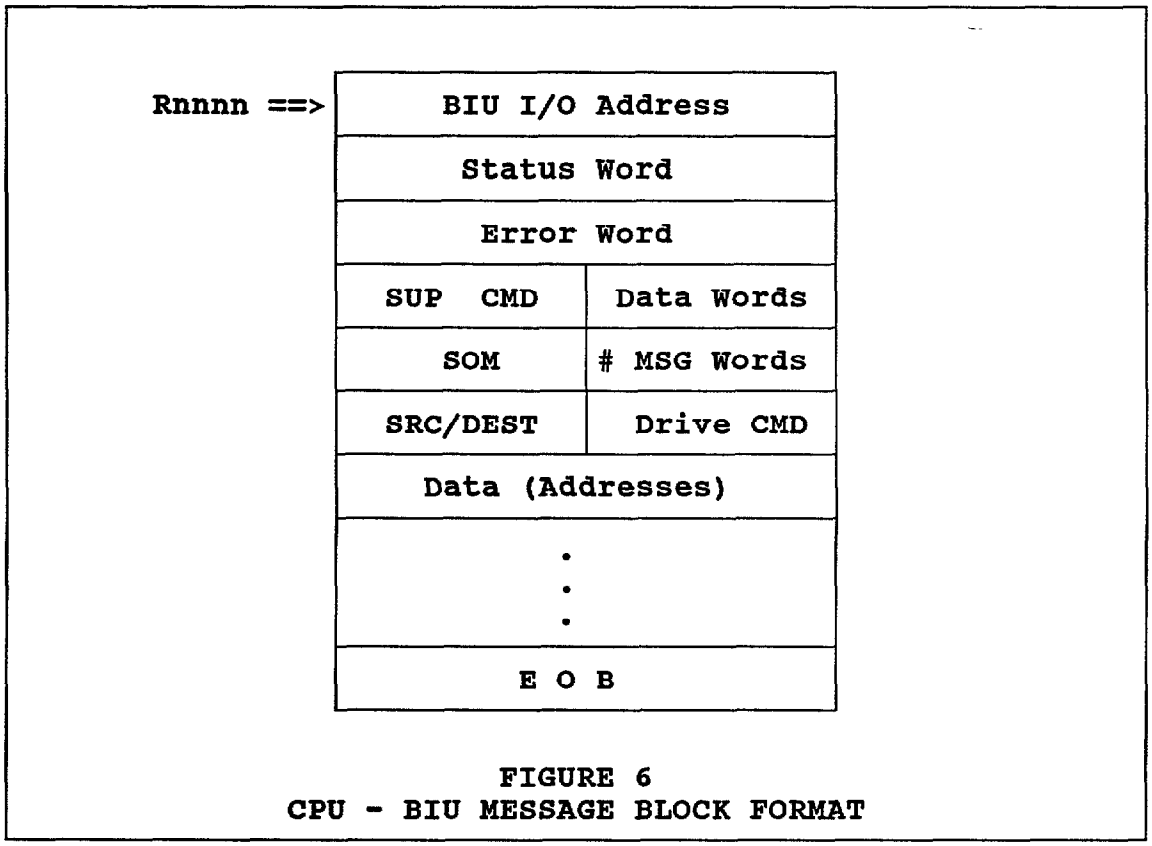


FIGURE 6
CPU - BIU MESSAGE BLOCK FORMAT

The format of the BIU message is similar to that used for BIU-LAN communications. Additional characters, Start Of Message (SOM), End Of Block (EOB), and Source/Destination (SRC/DEST) have been added to identify each BIU message within the data field. Multiple messages may be chained within the data field of the Command Block. Null characters may be inserted anywhere in the data field except between SOM and SRC/DEST.

DPREQ: The DPREQ is executed through ladder logic similar to that shown in FIGURE 7. Input contact Ixxxx (generally from a one-shot contained in ladder logic software) triggers the communication. Rnnnn refers to the first register in the message Command Block. The register contents (HHHH in Hexadecimal) correspond to the first I/O address selected for the BIU channel to be accessed (plus 1000 decimal).

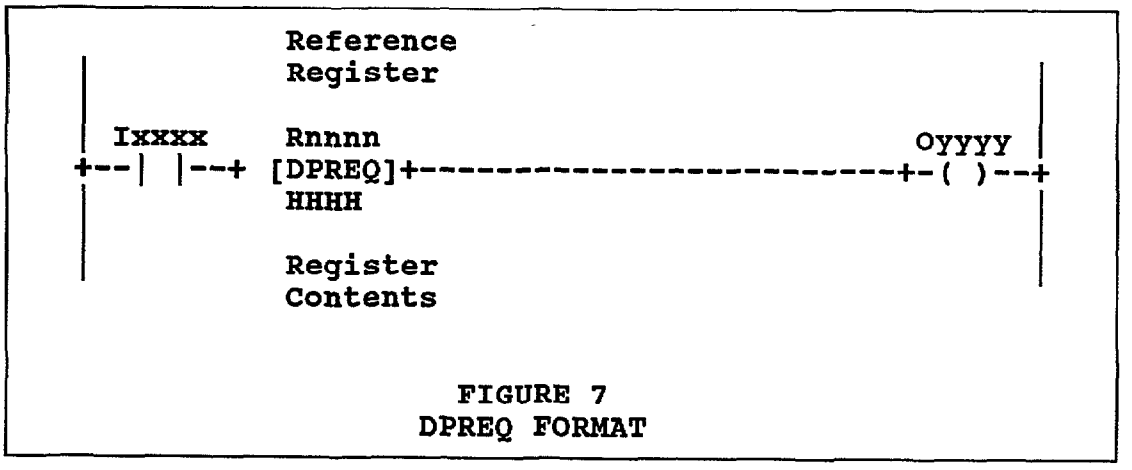


FIGURE 7
DPREQ FORMAT

When input Ixxx is closed (True), the communications window is opened as the line of logic is executed and the message block is sent to the BIU. If data is exchanged properly, there is no "power flow" through the DPREQ instruction and the window is considered successful. If the window is unsuccessful, there is "power flow" and output Oyyyy turns on.

DATA TRANSFER: The message Command Block is loaded with the necessary commands and data prior to execution of the DPREQ instruction. At the end of the window, the Status and Error words are updated and, if a read function was executed, information from the BIU is transferred to the data fields.

Status bits inform the PLC if the transmit buffer is full, when there are messages in the receive buffer, or when overwriting of the receive buffer occurs. Generally, a command is executed on every processor sweep that reads the BIU Status and Error words. This is done to assure that alarms are received without delay, even if no data is expected.

COMMAND PRIORTY: Certain commands are directed to the BIU and are not transmitted over the LAN. These "SUP Commands" are transmitted immediately upon execution to the SUP microprocessor. Other commands are transmitted by the BIU to members on the LAN. These "Drive Commands" are inserted into a queue to be formatted and transmitted at the end of the logic sweep. They are transmitted on a priority basis in which STOP commands have priority over all others issued on the same sweep.

SERIES SIX PLUS PLC WINDOW: The Series Six Plus controller provides an additional WINDOW command that may be used for communicating with the BIU. The format of this command is shown in FIGURE 8. The first three digits of the WINDOW address (0003) indicate the Series Six Plus channel number (000) and the fourth digit (3) provides the BIU card address from the formula:

1000 + (8 x Y = 1) = BIU ADDRESS

For our example: 1000 + (8 x 3 + 1) = 1025

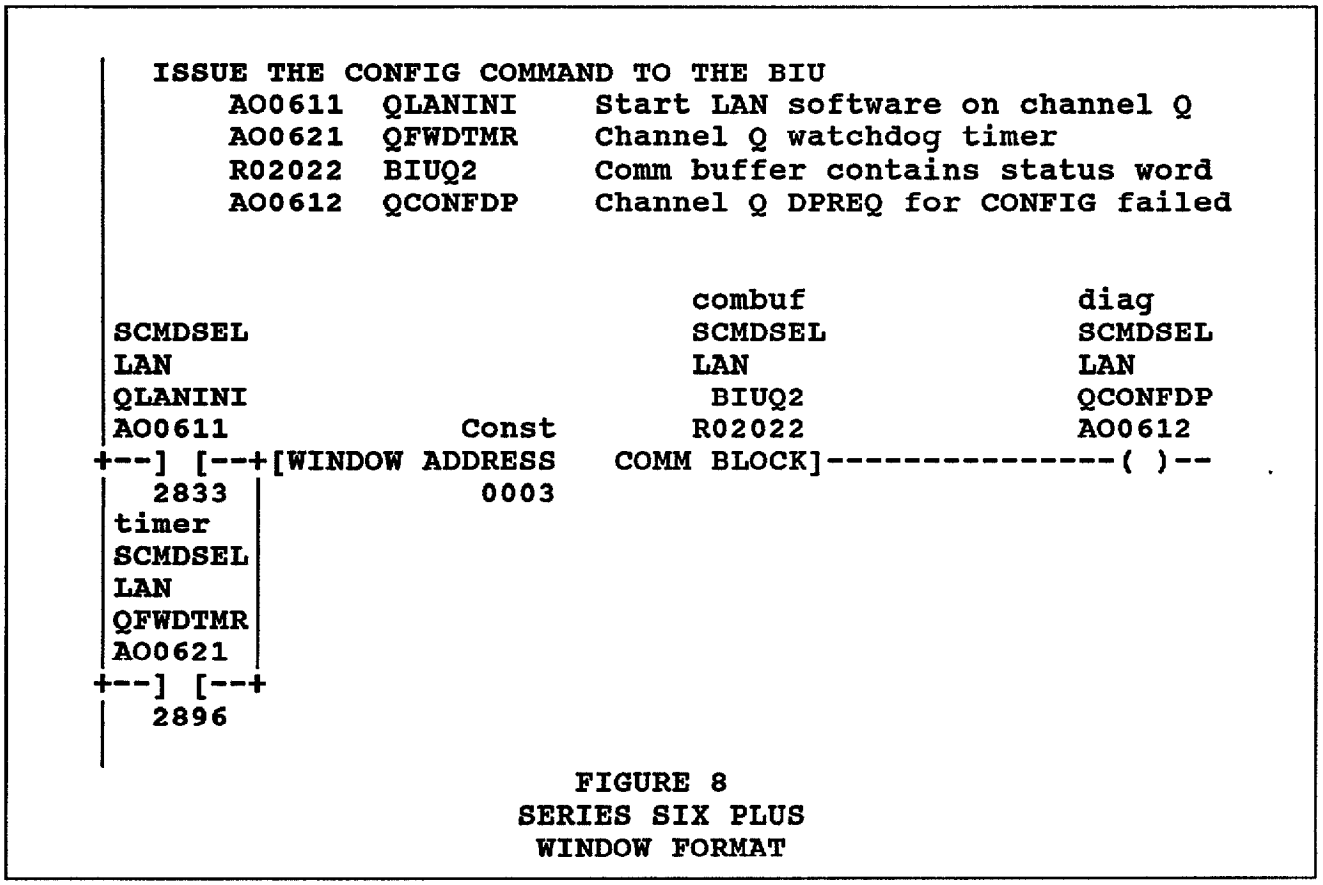


FIGURE 8
SERIES SIX PLUS
WINDOW FORMAT

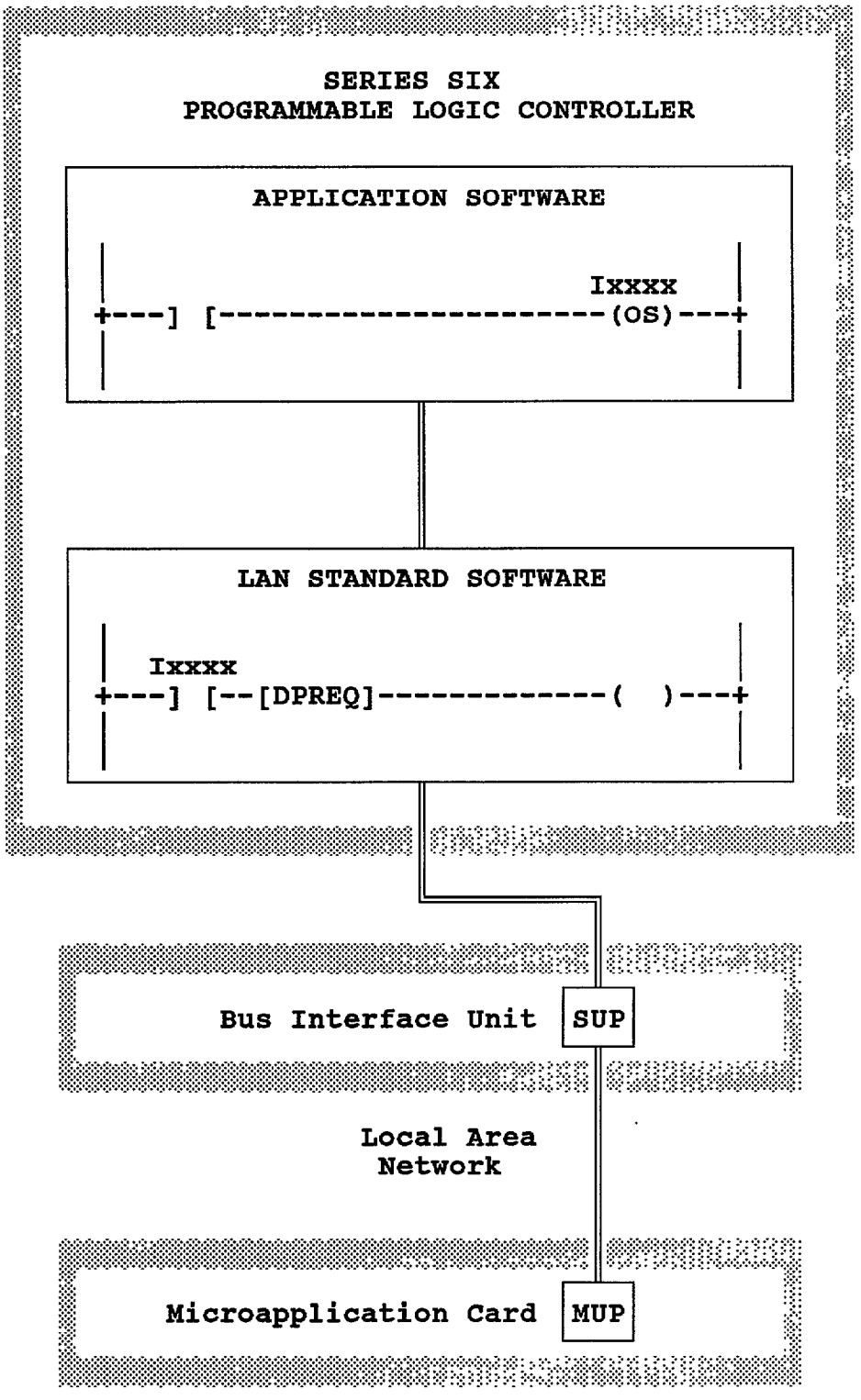


FIGURE 9
LAN HIERARCHY

4. SOFTWARE AND COMMANDS

The Bus Interface Unit and Microapplication cards each contain their own microprocessors and software in programmable read only memory (PROM) for LAN communications. The software for the programmable logic controller is programmed in ladder logic that is loaded into the Series Six PLC and divided into two portions:

A. **LAN Standard Software** - serves as the "operating system" for communications between the PLC and the BIU. It provides Command Block register formatting, DPREQ (or WINDOW) instructions, and associated code for interpretation and execution of LAN commands.

B. **Application Software** - is ladder logic code, consisting mostly of one-shots, that trigger LAN Standard Software codes to access LAN communications functions.

This software is custom designed by GE DRIVE SYSTEMS engineers, using high level programming aids, as part of the complete drive system package. Brief descriptions of the LAN Standard and Application Software follow. Various commands used in application code are listed for reference in troubleshooting and monitoring system operation.

CAUTION: The Local Area Network (LAN) and associated software are part of a coordinated drive system package. No changes to this code should be attempted except as approved by GE Drive Systems engineering. Unauthorized code changes may cause drive misoperation and will void the warranty.

LAN STANDARD SOFTWARE

As detailed in the Communications Protocols paragraph, the Series Six PLC communicates with the BIU through DPREQ and WINDOW instructions that are executed in ladder logic. Data required for this exchange must first be loaded into Command Blocks in register memory, that is then accessed as the window is executed. The LAN Standard Software provides the necessary instructions and register formatting used in accessing the LAN and performing the associated data handling functions. The LAN Standard Software may be functionally divided into three parts, as follows:

A. **Basic Functions:** Software to initialize and support both channels of the BIU and to monitor up to seven feedback and status parameters (including Fault Code) from the drives.
   1. Enable and configure the BIU
   2. Teach the BIU the number of feedback variables expected for each drop.
   3. Issue FREEZE commands to retrieve data from the drives.
   4. Transfer FREEZE data from the drives to register memory.
   5. Transfer Status and Error words to register memory.

B. **Referencing Functions:** Software to format drive commands, combining a destination number with the command numbers, to allow a variety of commands to be issued to specific drives according to Drop Number, Group Number, or Zone.

C. **Advanced Diagnostic Functions:** Software for more precise network monitoring and drive history reporting. Provides additional troubleshooting commands to address the LAN and drives.
   1. Transfer BIU configuration parameters to registers in the PLC and allow inspection of the Link Monitor showing which drops are active.
   2. Read SUP revision numbers for identification of BIU software level.
   3. Drive History collection and reporting accesses a 51 frame circular file containing "snapshots" of up to seven drive variables per frame (logged over the previous 280 milliseconds).

The LAN Standard Software package provided with a particular drive system is a subset of the total LAN Standard. It will always include the Basic Functions. Other functions are provided as options specified for each system.

APPLICATION SOFTWARE

The Application Software is ladder logic code which is designed specifically for the application. It directs system operation, loading data registers, and triggering one-shots that are interpreted by the LAN Standard Software to format and transmit LAN commands to the BIU.

Nicknames, defined in the LAN Standard, are used for register and command references for easy reference and coordination with the LAN Standard. Nicknames are selected to be meaningful, such as M6MDMUP for the MOTOR 6 MODIFY MUP command and M6MDMAD for the Motor 6 MODIFY MUP ADDRESS. Generalized references to commands are listed in these instructions with the variable portion replaced by a slash mark (i.e/MDUMP). Actual nicknames used in the code contain meaningful characters in place of the slash, such as "Q" or "P" for BIU channels Q or P and "M6" or "W3" for drives designated as Motor 6 or Winder 3.

NOTE: *LAN Standard Software functions and commands are incorporated into the system software as required and specified for each system. Not all commands listed are supported in every system. Additional custom commands may be provided on a particular system that are not listed in this publication.*

Following is a partial listing of some of the commands that are most frequently used in Application Software, including their structure and function. A summary of these commands is provided in the table in APPENDIX B of this instruction book.

BIU CONFIGURATION

/TRCBLK Transfer BIU Configuration Block

This command transfers seven data words, defining the current BIU confifuration for either channel P (PTRCBLK) or channel Q (QTRCBLK), to fixed areas of register memory. The seven data words are as follows:

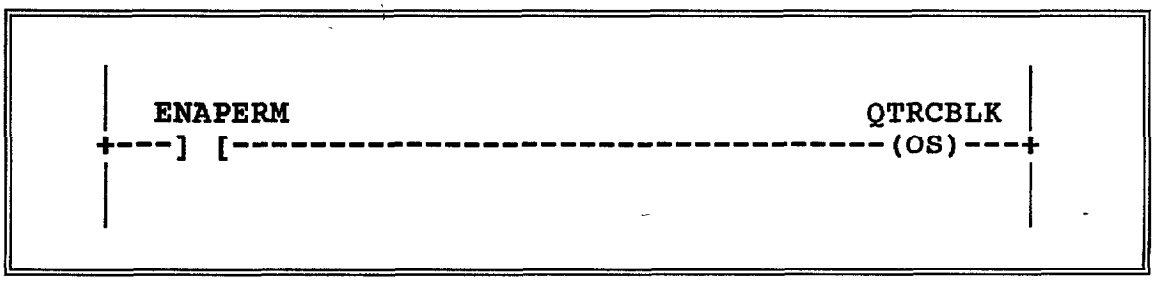
/CNSTAT	Status	
	Bit 0	Run Link Monitor periodically
	Bit 1	Run Link Monitor once
	Bit 2	Monitor polled
	Bit 3	Reserved
	Bit 4	57.6 Kilobaud
	Bit 5	Reserved
	Bit 6	CRT Mode
	Bit 7	BIU Protocol enabled
	Bit 8 to Bit 15	Reserved

/CNSTAT **Status (continued)**

/CNDROP	Drop Number for BIU
/CNGRUP	Group Number for BIU
/CNTOTD	Total number of drops on LAN channel
/CNLDRP	Drop Number of last drop to not acknowledge
/CNRSP1	Link Monitor response for drops 0-15. (Drop 0 is least significant bit) Bits set to "0" indicate active drops.
/CNRSP2	Link Monitor response for drops 16-31. (Drop 16 is least significant bit) Bits set to "0" indicate active drops.

Refer to the Diagnostics and Troubleshooting paragraphs for details on monitoring and interpreting the Configuration Block Data.

The following rung might be included in the application code to execute this command for channel Q of the LAN. The contact labelled ENAPERM represents any combination of series or parallel contacts used as an enable permissive.



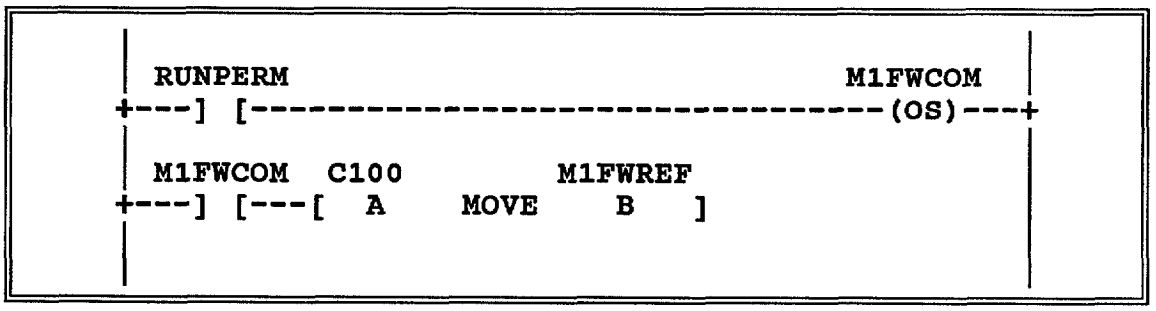
DRIVE RUN COMMANDS

There are a variety of instructions which may be provided to enable the PLC to command a drive to close its main contactor and to run in either forward or reverse. The reference at which the drive operates may be programmed over the LAN or predefined in any of several RAM addresses within the drive.

/FWCOM Run Forward At /FWREF

This commands the specified drive to run in the forward direction at the reference loaded in register /FWREF as scaled by drive parameter ADR068-REFSF. This command may be issued while the drive is running to change speeds during operation. The following example commands motor M1 (Motor 1) to run using a reference of 100 (scaled by drive parameter REFSF) when the contact labelled RUNPERM closes.

Example:



/RUNFO Run Forward Using RUNRF

This commands the drive to run in the forward direction using the regulation mode and reference defined in drive parameters ADR006-RUNJP and ADR061-RUNRF, respectively.

/RUNF1 Run Forward Using RUN1

This commands the drive to run in the forward direction using the regulation mode and reference defined in drive parameters ADR007-R1JP and ADR064-R1RF, respectively.

/RUNF2 Run Forward Using RUN2

This commands the drive to run in the forward direction using the regulation mode and reference defined in drive parameters ADR008-R2JP and ADR065-R2RF, respectively.

/RSLWF Run Forward Using SLOW

This commands the drive to run in the forward direction using the reference defined in drive parameter ADR062-SLWRF.

/RVCOM Run Reverse At /RVREF

This commands the specified drive to run in the reverse direction at the reference loaded in register /RVREF as scaled by drive parameter ADR068-REFSF. This command may be issued while the drive is running to change speeds during operation. Command execution is similar to that illustrated for the /FWCOM command.

NOTE: *Do not negate the reference when loading register /RVREF. The command already calls for reverse rotation of the motor.*

/RUNRO Run Reverse Using RUNRF

This commands the drive to run in the reverse direction using the regulation mode and reference defined in drive parameters ADR006-RUNJP and ADR061-RUNRF, respectively.

/RUNR1 Run Reverse Using RUN1

This commands the drive to run in reverse direction using the regulation mode and reference defined in drive parameters ADR007-R1JP and ADR064-R1RF, respectively.

/RUNR2 Run Reverse Using RUN2

This commands the drive to run in the reverse direction using the regulation mode and reference defined in drive parameters ADR008-R2JP and ADR065-R2RF, respectively.

/RSLWR Run Reverse Using SLOW

This commands the drive to run in the reverse direction using the reference defined in drive parameter ADR062-SLWRF.

/R+REF Run At Last REF _REG

This commands the drive to run at the reference last defined in the drive Reference Register (RAM address 510).

/R-REF Negate REF_REG And Run

This commands the drive to reverse the sign of the reference last defined in the drive Reference Register (RAM address 510) and run.

DRIVE STOP AND RESET COMMANDS

/CSTOP Open Contactor And Coast

This commands the drive to open its main contactor and coast.

/STOP Stop Per Preset

This commands the drive to stop in a preset manner that is defined by drive parameter ADR004-STPJP.

/QSTOP Quick Regenerative Stop

This commands the drive to use regenerative control and bypass linear time to stop as quickly as allowed by the current limit setting.

/TRIP Trip Stop-Await Reset

This commands the drive to open its main contactor and coast. Either a hardware reset from the drive control card or a soft reset via LAN command /RESET is required before the drive may be restarted.

/RESET Reset Drive

This is a "soft" reset which resets most fault conditions in the drive but does not upload the drive EEROM to overwrite the contents of the drive's RAM.

DRIVE JOG COMMANDS

/JOGF Run Forward Using JOGRF

This commands the drive to close its main contactor and run in the forward direction using the reference defined in drive parameter ADR063-JOGRF. The/JGSTP command is used to stop jogging. This command may be used for slack take up and issued while the drive is running to add JOGRF to the speed reference. The/JGSTP command instructs the drive to return to the original reference.

/JOGR Run Reverse Using JOGRF

This commands the drive to close its main contactor and run in the reverse direction using the reference defined in drive parameter ADR063-JOGRF. The /JGSTP command is used to stop jogging. This command may be used for slack take down and issued while the drive is running to subtract JOGRF from the speed reference. The /JGSTP command instructs the drive to return to the original reference.

/JGSTP Stop Jogging

This commands the drive to stop jogging, cancelling the forward (/JOGF) or reverse (/JOGR) jog commands.

DRIVE INCREMENT/DECREMENT COMMANDS

/ -SPD Start Decrementing Speed

This commands the drive to decrease speed reference until the drive reaches zero speed (or maximun negative speed) or until the /STSPD command is received.

/+SPD Start Incrementing Speed

This commands the drive to increase speed reference until the drive reaches maximum positive speed or until the / STSPD command is received.

/STSPD Stop Incr / Decr Speed

This commands the drive to stop speed increase or decrease (deactivates / + SPD or /-SPD) and to hold at the last speed reached.

/-DRAW Decrement Draw Ratio

This commands the drive to decrement the draw ratio (in RUP RAM) by the value contained in drive parameter ADR060-RATIS. The ratio will remain at the updated value until changed by a /-DRAW, /+DRAW or a modify RAM command. A hard reset returns the ratio to 100% of the value stored in drive parameter ADR059-RATIO

/+DRAW Increment Draw Ratio

This commands the drive to increment the draw ratio (in RUP RAM) by the value contained in drive parameter ADR060-RATIS. The ratio will remain at the updated value until changed by a /-DRAW, /+DRAW or a modify RAM command. A hard reset returns the ratio to 100% of the value stored in drive parameter ADR059-RATIO

DRIVE MODE

/MDMAN Change Mode To Manual

This commands the drive to switch to the manual regulation mode.

/MDATO Change Mode To Auto

This commands the drive to switch to the auto regulation mode.

/MDSEL Mode Select By /MDDAT
/MDSL1 Mode Select By /MDDT1
/MDSL2 Mode Select By /MDDT2
/MDSL3 Mode Select By /MDDT3
/MDSL4 Mode Select By /MDDT4
/MDSL5 Mode Select By /MDDT5

These command the drive to select predefined operating modes according to the bit settings of the associated data word. Since each of the six Mode Select commands can select multiple modes, it is not necessary to issue more than one such command per sweep.

/MDD_Bit Pattern	Dec. Wt.	Function
....1	1	Enable second byte
....1.	2	Enable diameter hold
.... .1..	4	Inhibit position loop
.... 1...	8	Enable field economy
....1	16	Enable FLDM2
....1.	32	Enable FLDM3
.... .1..	64	nhibit regeneration
.... 0...		(spare bit)
....1	256	Enable RUNO
.... ..1.	512	Enable RUN1
.... .1..	1024	Enable RUN2
.... 1...	2048	Enable SLOW
...1	4096	Enable REVERSE
000.		(spare bits)

/MDSMD Modify SP Mode /MDSDT

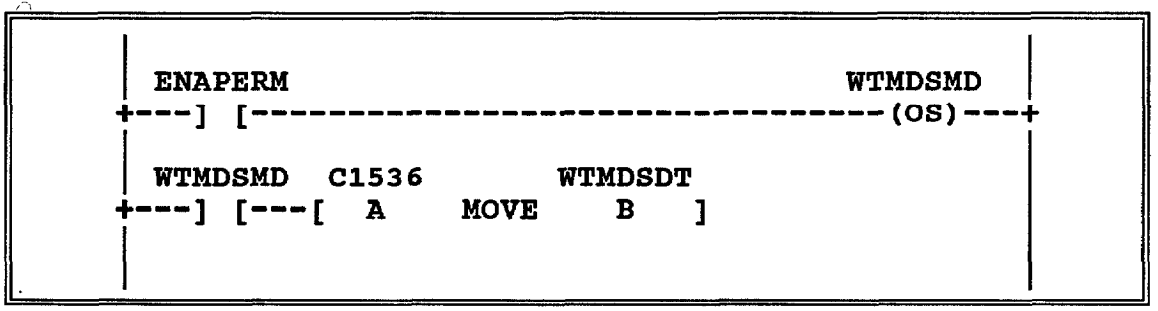
This commands the drive to select predefined operating modes according to the bit settings of the /MDSDT data word.

/MDSDT Bit Pattern	Dec. Wt.	Function
.... ..00	0	Select Main Motor gains (ADR015 through ADR041)
.... ..01	256	Select Motor 1 gains (ADR143 through ADR154
.... ..10	512	Select Motor 2 gains (ADR155 through ADR166)
.... ..11	768	Select Motor 3 gains (ADR167 through 178)
.... .1..	1024	Enable IFOLLOW (enter Current Follower mode)
.... 1...	2048	Enable RECAL TACH (reduce speed regulator gain by 8:1)
...1	4096	Enable RCAL REF (multiply drive reference by SLODN/256)
..1.	8192	Enable RECAL ILIM (use ILMRC instead of ILMMX for current limit)
....1	1	Enable REF INHIBIT (clamp drive reference to zero, regardless of the source reference)
....1.	2	Enable REV INHIBIT (clamp drive reference to zero if the source reference is negative)
.... .1..	4	Enable FWD INHIBIT (clamp drive reference to zero if the source reference is positive)
.... 1...	8	Enable LOW GEAR (switch to increased Position Loop is active)
00.. .... 0000		(spare bits)

The following rung might be included in the application code to enable the Current Follower mode and select Motor 2 gains for a drive, identified as "WT", on a wire turning roll. The constant (C1536) is calculated from the sum of the decimal weights of the bits for the selected functions.

1536 = 512 (select Motor 2 gain) + 1024 (enable IFOLLOW)

The contact labelled ENAPEM represents any combination of series or parallel contacts used as enable permissive.



DRIVE MUP MODIFY

There are three commands which may be provided to allow the downloading of operating data or "recipes" to the drive from the Series Six Programmable Logic Controller. The Microapplication card's microprocessor (MUP) uses random access memory (RAM) for storage of operating data in two pages (0 and 1) that are mirror images. The recipe download procedure involves the following command sequence:

1. **Hold** PAGE 0 for setup by issuing the /HDSET command to transfer MUP reference from PAGE 0 to PAGE 1.
2. **Modify** MUP RAM, downloading the new recipe data to PAGE 0 by issuing a number of "Modify MUP" commands (i.e. /MDMUP, /MDMP1.../MDMP5).
3. **Execute** the new recipe by issuing the /EXSET command to transfer MUP reference back to PAGE 0.

/HDSET Hold Page 0 for setup

This commands the drive to transfer MUP reference from PAGE 0 to PAGE 1 in preparation for modifying MUP RAM.

		ADDR	LSBD	MSBD
/MDMUP	Modify MUP	**/MDMAD** to	**/MDMDT**	**/MDMDM**
/MDMP1	Modify MUP	**/MDMA1** to	**/MDMD1**	**/MDMM1**
/MDMP2	Modify MUP	**/MDMA2** to	**/MDMD2**	**/MDMM2**
/MDMP3	Modify MUP	**/MDMA3** to	**/MDMD3**	**/MDMM3**
/MDMP4	Modify MUP	**/MDMA4** to	**/MDMD4**	**/MDMM4**
/MDMP5	Modify MUP	**/MDMA5** to	**/MDMD5**	**/MDMM5**

These command the drive to download recipe data to the indicated MUP RAM address (/MDMAD, /MDMA1.../MDMA5). Data which requires only two bytes is contained in one data register (/MDMDT, /MDMD1.../MDMD5). Where four bytes are required for double precision, the data is contained in two data registers, the least significant two bytes (/MDMDT, /MDMD1 ... /MDMD5) and the most significant two bytes (/MDMDM, /MDMM1 ... / MDMM5). The most significant bit of the MUP address register (/MDMA_) is set to indicate that four byte data is being downloaded.

The six Modify MUP commands are executed in order of priority from /MDMUP to /MDMP5. If all six Modify MUP commands were issued during the same processor sweep, /MDMUP would execute first and /MDMP5 would execute last.

/EXSET Execute PAGE 0 Setup

This commands the drive to transfer operation to PAGE 0 and begin executing the new operating recipe.

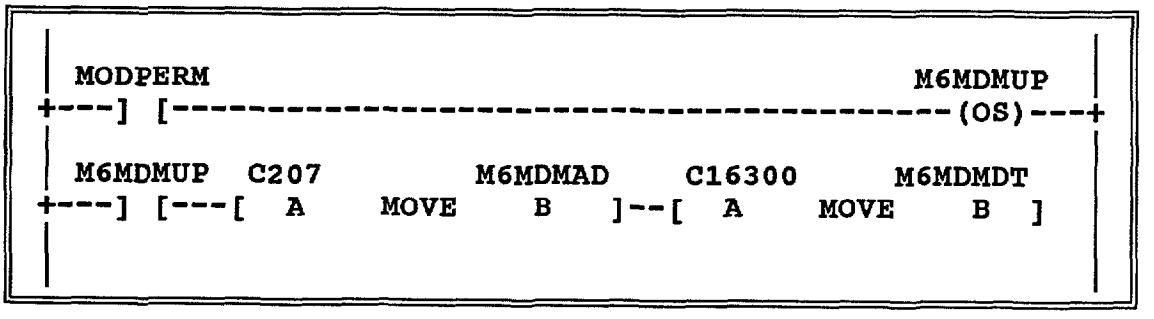
Following are two examples of Application Software ladder logic which might be used to download single precision (two byte) or double precision (four byte) data using the Modify MUP command.

Example: Two Byte Modify MUP

This example loads MUP RAM address 00CF hex (PRGMR1) of Motor 6 (M6) with the number 16300.

--- The contact labelled MODPERM represents any combination of series or parallel permissives.

--- A contact from oneshot M6MDMUP loads the MUP address (00CF hex = 207 dec) into M6MDMAD and the data (16300) into M6MDMDT.



Example: Four Byte Modify MUP

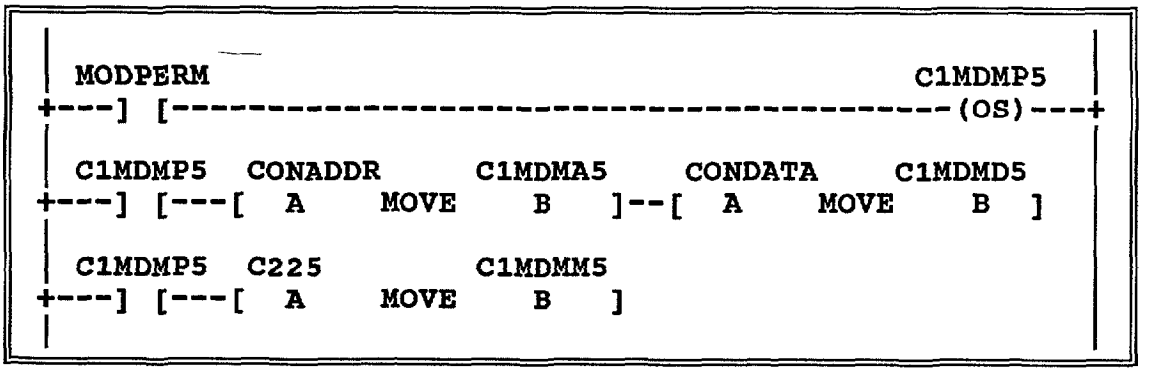
This example loads MUP RAM address 8310 hex (PC REF) of a drive designated as Conveyor 1 (C1) with the double precision number 16777215.

--- The contact labelled MODPERM represents any combination of series or parallel permissives.

--- A contact from oneshot C1MDMP5 loads the MUP address (33552) from CONADDR into C1MDMA5 and the least significant word of the data (65535) from CONDATA into C1MDMD5.

--- The second contact loads the most significant word (225) of data into C1MDMM5.

--- Since the maximum constant allowed by the Series Six PLC is C16383, registers CONADDR and CONDATA are used and it is assumed that they have been initialized to 33552 and 65535, respectively.



DRIVE RUP MODIFY COMMANDS

		ADDR	DATA
/MDRAM	Modify RUP	**/MDRAD** to	**/MDRDT**
/MDRM1	Modify RUP	**/MDRA1** to	**/MDRD1**
/MDRM2	Modify RUP	**/MDRA2** to	**/MDRD2**
/MDRM3	Modify RUP	**/MDRA3** to	**/MDRD3**
/MDRM4	Modify RUP	**/MDRA4** to	**/MDRD4**
/MDRM5	Modify RUP	**/MDRA5** to	**/MDRD5**

These command the drive to downlaod data (/MDRDT, /MDRD1 ... /MDRD5) to the indicated RUP RAM address (/MDRAD, /MDRA1... /MDRA5. The six Modify RUP commands are executed in order of priority from /MDRAM to /MDRM5. If all six Modify RUP commands were issued during the same processor sweep, /MDRAM would execute first and /MDRM5 would execute last. Application code to execute this command would use a oneshot (i.e. /MDRAM) with contacts triggering data moves to load registers with the RUP RAM address (i.e. /MDRAD) and the new RUP data (i.e. /MDRDT). This command is independent of drive operation and may be issued at any time.

DRIVE HISTORY

An option is available which records drive parameters in a circular history file, maintained in MUP, for uploading to the Series Six PLC. The file consists of 51 history "frames" (0-50) with each frame containing up to seven variables. The history is updated every two "firings" of the drive SCRs, providing a record of approximately 280 milliseconds (102 firings) span. History updating may be suspended to "freeze" history data preceding an event for uploading to the LAN. The event that freezes the history may be set to be any drive trip, a trigger by SLD4, or a LAN command.

There are **four** commands which may be provided to allow the uploading of historical operating data from the drive to the Series Six Programmable Logic Controller. The procedure involves the following command sequence:

1. Freeze the drive history by issuing the /FRHIS command to the drive to stop updating.
2. Request data from one of 51 history frames by issuing the /RQHIS command to the drive.
3. Transfer the contents of the history frame from SUP memory to Series Six registers by issuing the /HISTRY command to the BIU Channel (P or Q) serving the drive.
4. Reactivate the drive history by issuing the /REHIS to the drive to resume updating.

The REQUEST and TRANSFER cycle may be repeated, along with additional Series Six commands, to access and log all 51 frames of drive history. Where only one frame is required, it may be preferred to REACTIVATE the history before executing the TRANSFER to minimize the time that the history record is not updating.

/FRHIS Freeze History

This commands the specified drive to stop updating the history file, preserving the last recorded 51 frames for uploading.

/RQHIS Request History Frame /RQHAD

This commands the drive to upload the data contained in the history frame number specified by /RQHAD, where the contents of /RQHAD are defined as:

[/RQHAD] = 256 x the frame number

The data contained in the requested history frame is transferred to the SUP of the channel serving the addressed drive, pending transfer to predefined Series Six registers by the /HISTRY command.

/HISTRY Transfer History Frame

Transfers the history frame, previously uploaded by the /RQHST command, from the specified channel's SUP to ten continuous PLC registers as follows:

Register Name		Register Contents
Channel P	Channel Q	
PHSDRNO	QHSDRNO	Drive Drop Number
PHSFRM	QHSFRM	History Frame
PHSARMI	QHSARMI	Armature Current
PHSARMY	QHSARMV	Armature Voltage
PHSSTAT	QHSSTAT	Drive RUP Status
PHSPGM1	QHHSPGM1	MUP Variable PRGRM1
PHSPGM2	QHSPGM2	MUP Variable PRGRM8
PSPARE1	QSPARE1	Spare
PSPARE2	QSPARE2	Spare
PSPARE3	QSPARE3	Spare

Subsequent history transfers will overwrite the data in these registers with that from the last requested frame. Additional code may be required to utilize the data from these registers before the next history request and transfer are issued.

/REHIS Reactivate History

This commands the specified drive to resume updating the history file. Updating may also be reactivated by a hardware reset of the drive.

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The register group shown below indicates the order of the history frame information from the drive. There are 10 continuous history registers for each frame from channel P and 10 continuous registers for channel Q. The register addresses will be pre-addressed; the specific register addresses shown here were chosen to help clarify the example and may vary from those actually assigned.

R0210	R0209	R0208	R0207	R0206	R0205	R0204	R0203	R0202	R0201
PSPARE3	PSPARE2	PSPARE1	PHSPGM8	PHSPGM1	PHSSTAT	PHSARMV	PHSARMI	PHSFRM	PSHDRNO
R0220	R0219	R0218	R0217	R0216	R0215	R0214	R0213	R0212	R0211
QSPARE 3	QSPARE2	QSPARE1	QHSPGM8	QHSPGM1	QHSSTAT	QHSARMV	QHSARMI	QHSFRM	QSHDRNO

DRIVE FEEDBACK VARIABLES

Feedback information from the drives is returned to the BIU through a Freeze command issued by the LAN Standard Software. Each drive can be programmed to return up to 6 drive parameters plus the fault code. Refer to drive ADR215-MSET5 for selected feedback parameters. The freeze data is then transferred from the BIU to the Series Six register memory. This freeze/read sequence happens automatically.

The freeze rate can be adjusted from once per Series Six sweep to once every 10 seconds. Careful consideration must be given to the selection of the freeze rate. For guidelines, refer to the Data Flow Timing paragraph of this instruction book.

A watchdog timer provides a method of sensing that a drive on the LAN did not respond to a freeze request and reinitiates the freeze/read sequence. The freeze and watchdog timer preset values are predefined. For more detailed information on watchdog timer failure diagnostics refer to the Diagnostics and Troubleshooting paragraphs in this instruction book.

The drive feedback buffer is located at register location DM/1 (where DM = decoded matrix and / refers to P or Q channel). The length of the decoded matrix buffer is the total number of feedback variables from all drives + 2. This buffer location is dedicated to the decoded matrix and is only over-written by subsequent read commands. The contents of the buffer registers are unscaled drive feedback values, scaling is done in application code external to the LAN Standard. The buffer is filled in from location 1 to the end with the lowest to highest numbered drop's feedback parameters (see the Decoded Matrix example below). The maximum number of feedback variables from all drops is limited to 125 words (actual memory size is 127 words, 2 words are reserved for the freshness bits).

READ/4 (where / refers to P or Q channel) can be used as a pointer to extract items from the buffer. Since READ/4 will be over-written when a read command is issued, the pointer value must be restored before extracting buffer values.

EXAMPLE OF DECODED MATRIX BUFFER:

This example shows the ordering of information in the decoded matrix buffer after a READ command has been issued. There are 2 channels (P and Q) with 3 drops on each channel.

Channel P drop numbers and number of feedback parameters are:

First drop = drop number 1 with 7 parameters
Second drop = drop number 2 with 2 parameters
Third drop = drop number 3 with 3 parameters

Channel Q drop number and number of feedback parameters are:

First drop = drop number 1 with 3 parameters
Second drop = drop number 3 with 3 parameters
Third drop = drop number 5 with 3 parameters

The decoded matrix buffer for channel P is filled in with the following information:

DMP1= parameter 1 from drop number 1
DMP2 = parameter 2 from drop number 1
DMP3 = parameter 3 from drop number 1
DMP4 = parameter 4 from drop number 1
DMP5 = parameter 5 from drop number 1
DMP6 = parameter 6 from drop number 1
DMP7 = parameter 7 from drop number 1 (fault code)
DMP8 = parameter 1 from drop number 2
DMP9 = parameter 2 from drop number 2 (fault code)
DMP10 = parameter 1 from drop number 3
DMP11 = parameter 2 from drop number 3
DMP12 = parameter 3 from drop number 3 (fault code)
DMP13 = freshness bits for drops 16 through 32 (drop 16 = least significant bit)
DMP14 = freshness bits for drops 0 through 15 (drop 0 = least significant bit)

The decoded matrix buffer for channel Q is filled in with the following information:

DMQ1= parameter 1 from drop number 1
DMQ2 = parameter 2 from drop number 1
DMQ3 = parameter 3 from drop number 1 (fault code)
DMQ4 = parameter 1 from drop number 2 (dummy drop)
DMQ5 = parameter 1 from drop number 3
DMQ6 = parameter 2 from drop number 3
DMQ7 = parameter 3 from drop number 3 (fault code)
DMQ8 = parameter 1 from drop number 4 (dummy drop)
DMQ9 = parameter 1 from drop number 5
DMQ10 = parameter 2 from drop number 5
DMQ11 = parameter 3 from drop number 5 (fault code)
DMQ12 = freshness bits for drops 16 through 32 (drop 16 = least significant bit)
DMQ13 = freshness bits for drops 0 through 15 (drop 0 = least significant bit)

NOTE: *Channel Q drops 2 and 4 are dummy drops (not physically present), BIU convention allocates one register for fault code minimum.*

NOTE: *Register numbers for the nicknames are pre-assigned by the LAN Standard Software.*

5. STARTUP, DIAGNOSTICS AND TROUBLESHOOTING

This section describes the permissives necessary to initialize and run the LAN STANDARD SOFTWARE. It also points out and clarifies some areas that will help provide for a smooth startup. Also, this section outlines the LAN diagnostics and explains how to use these for troubleshooting.

PERMISSIVES AND INITIALIZATION

The /LANENA permissive (where / = P or Q channel) provides a method of delaying the initialization of the LAN software. This can be useful for delaying the LAN activity until some other software or hardware device is ready.

NOTE: *A coil with the /LANENA mnemonic must be energized in application code external to the LAN standard before the LAN software can initialize.*

COMMON STARTUP PROBLEM AREAS

Some of the most common LAN startup problems and solutions are documented in the following list. (Also refer to the "Diagnostics And Troubleshooting" paragraphs.)

1. **STATUS WORD = 0**
   Each BIU channel has a status word, Channel P status is mapped to P status and channel Q is mapped to Q status. The contents of this register should equal Axxx hex for a healthy, functioning channel. If the register contents are zero, check the BIU slot for an improper dip switch address setting. If P status or Q status are Bxxx hex, check drive EEROM. ADR214-MSET4 should equal number of drives plus 1 (for BIU) on channel.

2. **WATCHDOG TIMEOUTS**
   Each BIU channel has a watchdog failure counter word. Channel P is mapped to PWATERR and channel Q is mapped to QWATERR. If the contents of this register are incrementing, check that the number of words for each drop in the TEACH SUP COMMAND (82H) is equal to the number defined in ADR 215-MSET5 of the drive.

3. **DATA PROCESSING REQUEST FAILURES**
   Each BIU channel has a Data Processing Request (DPREQ) failure counter word. Channel P is mapped to PDPERR and channel Q is mapped to QDPERR. If the contents of this register is incrementing, there is an I/O addressing problem or a bad BIU card.

4. **BIU IS INITIALIZED BUT DROP WILL NOT RESPOND TO COMMAND**
   Check that drive ADR 010-APPJP has an odd value stored in it and ADR014-SPCJP has bit 7 set (contains 128 decimal). Also make sure LAN wires are terminated properly (see the Installation paragraphs of this instruction book).

DIAGNOSTICS AND TROUBLESHOOTING

The LAN diagnostics can be enabled with the following options:

LANDIAG enables supporting diagnostic source code for channels P and Q to be included.
PDIAG assigns 20 diagnostic registers for channel P.
QDIAG assigns 20 diagnostic registers for channel Q.

These options will allow a dedicated Series Six PLC register area to be filled in with diagnostic information for both channels of the BIU.

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The register group shown below is an example of the diagnostic variables available in the LAN STANDARD SOFTWARE. There are 20 continuous registers for channel P and 20 continuous registers for channel Q. The register addresses will be pre-assigned. The specific register addresses shown here were chosen to help clarify the example and may differ from those actually assigned.

R0170	R0169	R0168	R0167	R0166	R0165	R0164	R0163	R0162	R0161
PREV3	PREV2	PREV1	PCNRSP2	PCNRSP1	PCNCDRP	PCNTOTD	PCNGRUP	PCNDROP	PCNSTAT
R0180	R0179	R0178	R0177	R0176	R0175	R0174	R0173	R0172	R0171
PDIAGS4	PDIAGS3	PDIAGS2	PDIAGS1	PQUEERR	PDPERR	PWATERR	PINIERR	PERROR	PSTATUS
R0190	R0189	R0188	R0187	R0186	R0185	R0184	R0183	R0182	R0181
QREV3	QREV2	QREV1	QCNRSP2	QCNRSP1	QCNCDRP	QCNTOTD	QCNGRUP	QCNDROP	QCNSTAT
R0200	R0199	R0198	R0197	R0196	R0195	R0194	R0193	R0192	R0191
QDIAGS4	QDIAGS3	QDIAGS2	QDIAGS1	QQUEERR	QDPERR	QWATERR	QINIERR	QERROR	QSTATUS

DISPLAY SETUP AND INTERPRETATION

The following paragraphs describe the contents of the diagnostic registers and steps necessary to fill in the registers. Other options and commands required to fill in the registers will be described in each register section.

DIAGNOSTIC REGISTERS PCNSTAT - PCNRSP2

These seven registers display the result of SUP command "Transfer BIU Configuration Block for Channel P".

Mnemonic	Description
PCNSTAT	Channel P status word (see BIU command /TRCBLK)
PCNDROP	Channel P drop number
PCNGRUP	Channel P group number
PCNTOTD	Channel P total drops (number of drives + 1)
PCNLDRP	Channel P last drop not acknowledging
PCNRSP1	Channel P link presence bits for drops 0-15
PCNRSP2	Channel P link presence bits for drops 16-31

PCNSTAT (Channel P Configuration Status)

The seven bits shown are mapped into the high byte of PCNSTAT; typically, bits 7, 4, and 0 are set in the high byte.

Bit 7	BIU protocol enabled
Bit 6	CRT mode
Bit 5	Reserved
Bit 4	57.6K baud
Bit 3	Reserved
Bit 2	Monitor polled
Bit 1	Run link monitor once
Bit 0	Run link monitor periodically

PCNDROP (Channel P Configuration BIU Drop Number)
This is the drop number of the BIU for channel P; the LAN standard code will set this word to zero.

PCNTOTD (Channel P Configuration Total Drops)
This is the total number of drops that the BIU expects are on channel P. Typically this number is set to the number of drives on the channel plus one for the BIU card. Some systems may add another drop for the GPBIU (used for testing on the factory floor). The important thing to remember is that this value should be equivalent to the value in EEPROM address 214 for all drops on channel P of the LAN. If these numbers are not equal, the BIU will disable itself (see bit 12 of the status word). Change address 214 at the drive to match the total drops in the Series Six register as needed. DO NOT CHANGE THE SERIES SIX CODE.

PCNLDRP (Channel P Configuration Last Drop to Respond).
This is the last drop that did not acknowledge a LAN message. The contents of this register will be 00FF hex for an operational LAN. Values other than 00FF hex indicate the specific drop number that is having problems. Typically the problem is a loose connector on the drop's Microapplication card or a failed Microapplication card.

PCNRSP1 (Channel P Configuration Link Presence Bits)
This is a bit status word that indicates the status of drops 0 thru 15 on channel P (drop 0 is the least significant bit). A bit weight of zero indicates that the drop was operational the last time it received the network token. If the bit weights are one (for physical drops on the LAN), then check for loose LAN connectors and loose wires at those drops. If the problem still persists replace the Microapplication card.

PCNRSP2 (Channel P Configuration Link Presence Bits)
This is a bit status word that indicates the status of drops 16 thru 32 on channel P (drop 16 is the least significant bit). A bit weight of zero indicates that the drop was operational the last time it received the network token. If the bit weights are one (for physical drops on the LAN), then check for loose LAN connectors and loose wires at those drops. If the problem still persists replace the Microapplication card.

DIAGNOSTIC REGISTERS PREV1 - PREV 3

These three registers display the result of SUP command 0AH (Transfer SUP Revision Numbers) for channel P.

Mnemonic	Description
PREV1	Channel P SUP revision word 1
PREV2	Channel P SUP revision word 2
PREV3	Channel P SUP revision word 3

DIAGNOSTIC REGISTERS PSTATUS AND PERROR

These two registers display the result of SUP command Return Status and Error for channel P.

Mnemonic	Description
PSTATUS	Channel P status word
PERROR	Channel P error word

P CHANNEL "ACCUMULATOR" DIAGNOSTIC REGISTERS

The following four registers are accumulators of counters that monitor various functions of the LAN. Data will be mapped into these registers when option PDIAG is enabled.

Mnemonic	Description
PINIERR	Channel P initialization failures
PWATERR	Channel P watchdog timeouts
PDPERR	Channel P DPREQ failures
PQUEERR	Channel P drive command queue overflows

PINIERR, Initialization Errors - indicates the number of times the BIU card failed initialization. A non-zero value in PINIERR typically indicates an I/O addressing problem, a defective Bus Interface Unit (BIU) card, or an improper BIU card address on the Series Six backplane switches. (See the "Series Six Module Backplane DIP Switch Settings", Table 1 of this instruction book.)

PWATERR, Watchdog Timer Errors - indicates the number of times a drop (or drops) did not respond to the Freeze Request command. PWATERR increments by 1 each time a drop does not respond. If PWATERR is incrementing check that ADR215-MSET5 matches what the BIU was taught via the SUP teach command (82H) for each drop. The drive Microapplication (MUP) card or the LAN interface (LUP) card could also be defective.

NOTE: *Certain applications will shut drives off as part of a normal operating sequence causing PWATERR to increment. For applications that require* ***all*** *drives to be operational* ***all*** *of the time, PWATERR can be monitored for a non-zero value to indicate a critical drive communications problem.*

PDPERR - indicates the number of times the Series Six Data Processor Request (DPREQ) or window request (WINDOW) could not be properly opened or closed when requested. A non-zero value in PDPERR typically indicates an I/O addressing problem, a defective Bus Interface Unit (BIU) card, or an improper BIU card address on the Series Six backplane switches. (See the "Series Six Module Backplane DIP Switch Settings", Table 1 of this instruction book.)

PQUEERR - indicates the number of times the drive command queue has overflowed. PQUEERR should always be zero. If it is not, then too many drive commands have been inserted into the drive command queue for a given Series Six sweep, thus causing the queue to overflow.

DIAGNOSTIC REGISTERS PDIAGS1 - PDIAGS4

These diagnostic registers have been set aside as spares for channel P.

Mnemonic	Description
PDIAGS1	Channel P LAN diagnostic spare
PDIAGS2	Channel P LAN diagnostic spare
PDIAGS3	Channel P LAN diagnostic spare
PDIAGS4	Channel P LAN diagnostic spare

DIAGNOSTIC REGISTERS QCNSTAT - QCNRSP1

These seven registers display the result of SUP command "Transfer BIU Configuration Block for Channel Q".

Mnemonic	Description
QCNSTAT	Channel Q status word (see BIU command /TRCBLK)
QCNDROP	Channel Q BIU drop number
QCNGRUP	Channel Q BIU group number
QCNTOTD	Channel Q total drops on LAN (number of drives +1)
QCNLDRP	Channel Q last drop not acknowledging
QCNRSP1	Channel Q link pressure bits for drops 0-15
QCNRSP2	Channel Q link presence bits for drops 16-31

QCNSTAT (Channel Q configuration status).

The seven bits shown are mapped into the high byte of QCNSTAT; typically, bits 7, 4, and 0 are set in the high byte.

Bit 7	BIU protocol enabled
Bit 6	CRT mode
Bit 5	Reserved
Bit 4	57.6K baud
Bit 3	Reserved
Bit 2	Monitor polled
Bit 1	Run link monitor once
Bit 0	Run link monitor periodically

QCNDROP (Channel Q Configuration BIU Drop Number).

This is the drop number of the BIU for channel Q. The LAN standard code will set this word to zero.

QCNGRUP (Channel Q Configuration BIU Group Number).

This is the group number of the BIU for channel Q. The LAN standard code will set this word to zero.

QCNTOTD (Channel Q Configuration Total Drops).

This is the total number of drops that the BIU expects are on channel Q. Typically this number is set to the number of drives on the channel plus one for the BIU card. Some systems may add another drop for the GPBIU (used for testing on the factory floor). The important thing to remember is that this value should be equivalent to the value in EEPROM address 214 for all drops on channel Q of the LAN. If these numbers are not equal, the BIU will disable itself (see bit 12 of the status words). Change address 214 at the drive to match the total drops in the Series Six register as needed. DO NOT CHANGE THE SERIES SIX CODE.

QCNLDRP (Channel Q Configuration Last Drop To Respond).

This is the last drop that did not acknowledge a LAN message. The contents of this register will be 00FF hex for an operational LAN. Values other than 00FF hex indicate the specific drop number that is having problems. Typically the problem is a loose connector on the drop's Microapplication card or a bad Microapplication card.

QCNRSP1 (Channel Q Configuration Link Presence Bits).

This is a bit status word that indicates the status of drops 0 thru 15 on channel Q. Drop 0 is the least significant bit. A bit weight of zero indicates that the drop was operational the last time it received the network token. If the bits weights are one (for physical drops on the LAN), then check for loose LAN connectors and loose wires at those drops. If the problem still persists, replace the Microapplication card.

QCNRSP2 (Channel Q Configuration Link Presence Bits).
This is a bit status word that indicates the status of drops 16 thru 31 on channel Q. Drop 16 is the least significant bit. A bit weight of zero indicates that the drop was operational the last time it received the network token. If the bits weights are one (for physical drops on the LAN), then check for loose LAN connectors and loose wires at those drops. If the problem still persists replace the Microapplication card.

DIAGNOSTIC REGISTERS QREV1 - QREV3

These three registers display the result of SUP command "Transfer SUP Revision Numbers for channel Q".

Mnemonic	Description
QREV1	Channel Q SUP revision word 1
QREV2	Channel Q SUP revision word 2
QREV3	Channel Q SUP revision word 3

DIAGNOSTIC REGISTERS QSTATUS AND QERROR

These two registers display the result of SUP command "Return Status and Error for channel Q".

Mnemonic	Description
QSTATUS	Channel Q status word
QERROR	Channel Q error word

Q CHANNEL "ACCUMULATOR" DIAGNOSTIC REGISTERS

The following **four** registers are accumulators of counters that monitor various functions of the LAN. Data will be mapped into these registers when option QDIAG is provided.

Mnemonic	Description
QINIERR	Channel Q initialization failures
QWATERR	Channel Q watchdog timeouts
QDPERR	Channel Q DPREQ failures
QQUEERR	Channel Q drive command queue overflows

QINIERR, Initialization Errors - indicates the number of times the BIU card failed initialization. A non-zero value in QINIERR typically indicates an I/O addressing problem, a defective Bus Interface Unit (BIU) card or an improper BIU card address on the Series Six backplane switches. (See the "Series Six Module Backplane DIP Switch Settings", Table 1 of this instruction book.)

QWATERR, Watchdog Timer Errors - indicates the number of times a drop or drops did not respond to the Freeze Request command. QWATERR increments by 1 each time a drop does not respond. If QWATERR is incrementing, check that ADR215-MSET5 matches what the BIU was taught via SUP teach command (82H) for each drop. The drive Microapplication (MUP) card or the LAN interface (LUP) card could also be defective.

NOTE: *Certain applications will shut drives off as part of a normal operating sequence causing QWATERR to increment. For applications that require* ***all*** *drives to be operational* ***all*** *of the time, QWATERR can be monitored for a non-zero value to indicate a critical drive communications problem.*

QDPERR - indicates the number of times the Series Six Data Processor Request (DPREQ) or Window Request (WINDOW) could not be properly opened or closed when requested. A non-zero value in QDPERR typically indicates an I/O addressing problem, a defective Bus Interface Unit (BIU) card, an improper BIU card, or an improper BIU card address on the Series Six backplane switches. (See the "BIU Series Six PLC Backplane DIP Switch Settings", Table 1 of this instruction book.)

QQUEERR - indicates the number of times the drive command queue overflowed. QQUEERR should always be zero. If it is not, then too many drive commands have been inserted into the drive queue for a given Series Six sweep, thus causing the queue to overflow.

DIAGNOSTIC REGISTERS QDIAGS1 - QDIAGS2

These diagnostic registers have been set aside as spares for channel Q.

Mnemonic	Description
QDIAGS1	Channel Q LAN diagnostic spare
QDIAGS2	Channel Q LAN diagnostic spare
QDIAGS3	Channel Q LAN diagnostic spare
QDIAGS4	Channel Q LAN diagnostic spare

QUEUE DIAGNOSTICS

The following diagnostic mnemonics are generated by the LAN standard code and may be helpful when writing application code.

1. **/QUEFUL (where / = P or Q channel)**
   This mnemonic is associated with an auxiliary output address. The output will energize (logic 1) if too many drive commands have been issued in a Series Six sweep, thus causing the drive command queue to overflow. Any commands issued when this bit is active will NOT be transmitted over the LAN.

2. **/QUEEMT (where / = P or Q channel)**
   This mnemonic is associated with an auxiliary output address. The output will energize (logic 1) if the drive command queue is empty. This bit can be referenced after the PROCESS DRIVE COMMANDS subroutine to see if the subroutine formatted and transmitted all of the commands in the drive command queue. A logic 0 indicates that all commands were not processed on this sweep and will be transmitted on subsequent sweeps. The LAN is running at maximum efficiency if the queue is empty on return from the subroutine.

DATA FLOW TIMING

An approximate loop time of the LAN can be calculated by using the following formulas:

Total Loop Time for Drive Commands =
(Number of drops) x 750 usec. +
(Number of bytes transmitted) x 280 usec. -
(Number of drops transmitting x 750 usec.

Freeze Response Time =
[(Number of words returned from drive x 3) +7] x 280 usec

Number of Drops
The number of drops is the total number of drops on the channel, including the BIU.

Number of Drops Transmitted
The BIU is the master and is responsible for transmitting commands over the LAN, therefore when issuing commands from the BIU, the number of drops transmitted equal one. Feedback data is returned to the Series Six from the drives via a broadcast freeze command. Since the command is broadcast, all drives will respond and the number of drops transmitted will equal the number of drops on the channel.

Number of Bytes Transmitted

The number of bytes transmitted is a function of the command type and destination of the command. Examples of lengths for number of bytes transmitted follow.

NOTE: *The number of Data Words for specific LAN commands can be obtained from the Application Software paragraph of this manual.*

Examples: Data Word Commands versus Number of Bytes Transmitted

- --- **Zero Data Word commands** require no supporting data (Jog Forward/Reverse, Stop, Run Forward/Reverse at RUNO) and transmit 4 bytes if directed to a Zone, Group or Broadcast, and 7 bytes if directed to a Drop
- --- **One Data Word commands** require one supporting Data Word (Run Forward/Reverse at a reference, Request History with a frame number) and transmit 7 bytes if directed to a Zone, Group, or Broadcast, and 10 bytes if directed to a Drop.
- --- **Two Data Word commands** require two supporting data words (Modify RUP Ram with an address and data), and transmit 10 bytes if directed to a Zone, Group, or Broadcast, and 13 bytes if directed to a Drop.
- --- **Three Data Word commands** require three supporting data words. The only one that exists is Modify MUP Ram with an address and two data words. It transmits 13 bytes if directed to a Zone, Group, or Broadcast, and 16 bytes if directed to a Drop.
- --- **The FREEZE command** is a broadcast command and its response will cause feedback information to be returned from all drives. Use the Freeze Response time equation to calculate the time it takes for all drops to return their feedback variables to the BIU.

NOTE: *Knowing the frequency of the command transmission activity and the loop time for each command, an approximate worst case total LAN loop time can be calculated.*

Example: LAN Timing

LAN layout for example:

40 drops total

25 drops on channel P

15 drops on channel Q

All drops are returning 5 feedback words via freeze

125 feedback words from channel P

75 feedback words from channel Q

Freeze Response Time Calculation Using the Freeze Response Time equation:

Channel P time = [(125 x 3) + 7] x 280 microsec = **.107 msec**

Channel Q time = [(75 x 3) + 7] x 280 microsec = **.65 msec**

Loop Time for a Modify RAM command to a ZONE:

(The Modify RAM command to a zone is a 10 byte command.)

Channel P time = (25 x 750 usec) + (10 x 280 usec) - (1 x 750 usec) = **20 msec**

Channel Q time = (15 x 750 usec) + (10 x 280 USEC) - (1 x 750 USEC) = **13.3 msec**

6. PARTS AND SERVICE

RENEWAL PARTS AND SERVICE

Renewal parts information for this equipment is included in the CUSTOM INSTRUCTION BOOK provided for the drive system. The renewal parts list is generated specifically for the equipment furnished on each order and provides information on parts relative to the particular rating and options incorporated in the drive at the time of shipment. If the renewal parts information is missing, a replacement list may be obtained by contacting your local GE Sales Office or GE DRIVE SYSTEMS Authorized Distributor with the drive MODEL and REQUISITION numbers. Parts lists may also be obtained direct from GE DRIVE SYSTEMS Technical Services, (814) 875-2075.

IT IS RECOMMENDED THAT THE PARTS LIST BE REVIEWED AND A SET OF SPARE PARTS BE MAINTAINED TO MINIMIZE DOWN-TIME SHOULD REPAIRS BE REQUIRED.

Renewal parts may be obtained by contacting your local GE Sales Office or GE DRIVE SYSTEMS Authorized Distributor. Service may be obtained by contacting your area GE Industry Services Engineering Office.

WARRANTY PARTS AND SERVICE

1. Consult the **GE DRIVE SYSTEMS Terms and Conditions** for details of the product warranty, including warranty period and parts or service coverage.

2. Warranty replacement parts or service assistance may be obtained direct from the factory during regular working hours **(8 a.m. - 5 p.m.; M-F)** by calling:

   GE DRIVE SYSTEMS
   PRODUCT SERVICE ENGINEERING
   ERIE, PA.
   (814) 875-3851

3. An **EMERGENCY HOT-LINE** has been established by **GE DRIVE SYSTEMS** to provide 24 hour telephone answering service to our customers.

   GE DRIVE SYSTEMS
   EMERGENCY HOT-LINE
   (814) 875-3219

   For a **PROMPT RESPONSE THE NEXT WORKING DAY,** the answering service operator will record the required information and your **NAME, COMPANY and PHONE NUMBER.** In cases of emergency, where assistance must be obtained immediately, the operator may be requested to contact a Customer Service Representative (for parts) or a Product Service Engineer (for technical assistance) at home.

4. So that we may serve you more expediently, please have the following information ready whenever contacting the factory:

DRIVE SERIAL NUMBER & MODEL NUMBER
PART NUMBER & DESCRIPTION

5. **GE DRIVE SYSTEMS** may request that failed parts replaced under warranty be returned to the factory. A RETURN AUTHORIZATION tag or number is to be obtained and included with any parts being returned. When returning parts, pack them carefully to prevent damage in transit and ship to:

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS
1100 LAWRENCE PARKWAY
ERIE, PA 16531
ATTENTION: PARTS REPAIR AND EXCHANGE

APPENDIX A: LAN STATUS AND ERROR WORD

TABLE A - 1: LAN Status Word Bits

Bit	Mnemonic	Description
0	/UPDATD	Data received from all known drops
1	/HISUPD	History matrix updated
2	/LANST2	Status word bit 2 spare
3	/LANST3	Status word bit 3 spare
4	/LANST4	Status word bit 4 spare
5	/LANST5	Status word bit 5 spare
6	/LANST6	Status word bit 6 spare
7	/MATACT	Frozen data received into matrix
8	/BUFOVR	Receive buffer overwriting undecoded
9	/FULBUF	Link transmit buffer full
10	/RECACT	Receive buffer active (not empty)
11	/TRNACT	Transmit buffer active (not empty)
12	/DISABL	Communications disable
13	/LNKMON	Link monitor function selected
14	/RECOVR	Recovered from reset, needs
15	/SUCWIN	Successful window communications

NOTE: *The slash "/" is replaced with the appropiate drop or BIU channel indentifier.*

LAN ERROR WORDS PERROR AND QERROR (Where / = P or Q)

The error words for channel P and Q should ALWAYS contain zero (see bit 0 for exception). A non-zero error word typically indicates that an incorrectly formatted LAN message was transmitted on the LAN All messages generated by the LAN standard code have been tested for integrity. If a non-zero error word is detected then check that nicknames BIU/25 (where / = P or Q) are not being overwritten by logic other than the LAN standard logic. BIU/1 thru BIU/2 is a continuous register area that is timeshared by all commands that are transmitted on the LAN.

APPENDIX A: LAN STATUS AND ERROR WORD (continued)

TABLE A - 2: LAN Error Word Bits

Bit	Mnemonic	Description
0	/DRPFLD	Drop failed to acknowledge
1	/LNKERR	Link error collision detected
2	/SRTMSG	Short message received from link
3	/ROWOVR	Matrix received row overflow
4	/ROWUND	Matrix received row underflow
5	/HISOVR	History matrix overflow
6	/HISUND	History matrix underflow
7	/LANER7	Error word bit 7 spare
8	/TIMERR	Window communications timing error
9	/LANER9	Error word bit 9 spare
10	/CHKERR	Window communications checksum error
11	/INVR-W	Unknown read/write command specified
12	/INVI-0	I/O address error, command rejected
13	/INVCMD	Invalid command field format
14	/INVDAT	Invalid data field format
15	/LANERF	Error word bit 15 spare

NOTE: *The slash "/" is replaced with the appropiate drop or BIU channel indentifier.*

APPENDIX B: COMMAND SUMMARY LIST

NOTE: *The slash "/" is replaced with the appropriate drop or BIU channel indentifier.*

Input Mnemonic	Data 1 Mnemonic	Data 2 Mnemonic	Data 3 Mnemonic	Description
/FWCOM	/FWREF			Run forward at /FWREF (packed)
/RUNF0				Run forward using RUN reference
/RUNF1				Run forward using RUN 1 reference
/RUNF2				Run forward using RUN 2 reference
/RSLWF				Run forward using SLOW reference
/RVCOM	/RVREF			Run reverse at /RVREF (packed)
/RUNRO				Run reverse using RUN reference
/RUNR1				Run reverse using RUN 1 reference
/RUNR2				Run reverse using RUN 2 reference
/RSLWR				Run reverse using SLOW reference
/RNPRE				Run at preselected auto/manual mode reference
/R+REF				Run at last speed in reference register
/R-REF				Run at neg (last speed in reference register)
/CSTOP				Open contactor and coast
/STOP				Stop per preset means
/QSTOP				Regenerative stop, no linear time
/TRIP				Coast stop and wait for reset
/JOGF				Run forward using JOG reference
/JOGR				Run reverse using JOG reference
/JGSTP				Stop drive when jogging
/-SPD				Start decrementing present speed
/+SPD				Start incrementing present speed
/-DRAW				Start decrementing present draw
/+DRAW				Start incrementing present draw
/STSPD				Stop speed increase or decrease
/RESET				Reset drive
/MDATO				Change drive to auto mode
/MDMAN				Change drive to manual mode
/MDSEL	/MDDAT			Mode select (RUN0, RUN1, RUN2, SLOW, REV)
/MDSL1	/MDDT1			Mode select (RUN0, RUN1, RUN2, SLOW, REV)
/MDSL2	/MDDT2			Mode select (RUN0, RUN1, RUN2, SLOW, REV)
/MDSL3	/MDDT3			Mode select (RUN0, RUN1, RUN2, SLOW, REV)
/MDSL4	/MDDT4			Mode select (RUN0, RUN1, RUN2, SLOW, REV)
/MDSL5	/MDDT5			Mode select (RUN0, RUN1, RUN2, SLOW, REV)

APPENDIX B: COMMAND SUMMARY LIST (continued)

NOTE: *The slash "/" is replaced with the appropriate drop or BIU channel indentifier.*

Input Mnemonic	Data 1 Mnemonic	Data 2 Mnemonic	Data 3 Mnemonic	Description
/HDSET				Hold MUP page 0 for setup
/EXSET				Execute from MUP page 0
/MDMUP	/MDMAD	/MDMDT	/MDMDM	Modify MUP RAM location /MDMAD with /MDMDT (1sw) and /MDMD2 (msw)
/MDMP1	/MDMA1	/MDMD1	/MDMM1	Modify MUP RAM location
/MDMP2	/MDMA2	/MDMD2	/MDMM2	Modify MUP RAM location
/MDMP3	/MDMA3	/MDMD3	/MDMM3	Modify MUP RAM location
/MDMP4	/MDMA4	/MDMD4	/MDMM4	Modify MUP RAM location
/MDMP5	/MDMA5	/MDMD5	/MDMM5	Modify MUP RAM location
/MDSMD	/MDSDT			Modify SP mode command (MOTOR 1, MOTOR 2, RECAL, ILIM, REF INHIBIT, REV INHIBIT, FWD INHIBIT, LOW GEAR)
/MDRAM	/MDRAD	/MDRDT		Modify RUP RAM location /MDRAD with /MDRDT
/MDRM1	/MDRA1	/MDRD1		Modify RUP RAM location /MDRA1 with /MDRD1
/MDRM2	/MDRA2	/MDRD2		Modify RUP RAM location /MDRA2 wih /MDRD2
/MDRM3	/MDRA3	/MDRD3		Modify RUP RAM location /MDRA3 with /MDRD3
/MDRM4	/MDRA4	/MDRD4		Modify RUP RAM location /MDRA4 with /MDRD4
/MDRM5	/MDRA5	/MDRD5		Modify RUP RAM location /MDRA5 with /MDRD5
/FRHIS				Freeze history
/REHIS				Reactivate history
/RQHIS	/RQHAD			History request frame
/HISTRY				Transfer history frame
/TRCBLK				Transfer BIU configuration block

APPENDIX C: LAN RELATED DRIVE FAULT CODES

Diagnostic Readout	Fault Code	Name	Description/Causes/Remedies
100100	FLA0	MLNKR	Serial Link (product protocol) Fault BIU disabled (EE211 MSET1.8 set); receive buffer overflowed. --- Enable the BIU
100101	FLA1	MSTCK	Stack Space filled due to MUP Application Software. The stack has overflowed the specified stack space. --- The application can adjust the specified stack space. --- The application should use less stack for storage.
100110	FLA2	MEPRM	EEROM Checksum is incorrect --- Replace the EPROM (U33).
100111	FLA3	MIRAM	The Internal Processor RAM Failed --- Replace the processor (U30).
101000	FLA4	MERAM	The External RAM Chip (6116 or 6164) failed --- Replace the RAM chip (U49).
101001	FLA5	MITMR	The Internal Processor Timers Failed --- Replace the processor (U49).
101010	FLA6	MSRAM	Unable to Modify and read RAM on Port Expander Chip (8155) --- Replace the port expander chip (U37).
101011	FLA7	MFMSG	Link Fault, Unable to Send Messages Processor has not heard the last 128 bytes sent out on serial port. --- May not have JP16, 17, and 15 tied 1 - 2 on Microapplication card. --- May have too many drops with the termination resistors in the circuit. (JP18 & 19 tied 1 - 2, max = 5 drops with termination resistors in the circuit) --- May have another drop programmed with the same drop number. --- All drops may not be programmed for same link size (ADR214). --- The serial lines may be shorted. --- The driver (U35) or receiver (U34) may be defective. --- The serial port on the processor may be defective (U30). --- The internal timer on the processor may be defective (U30). --- If actively using the LAN, then JP13 must tie 1 - 2. --- Replace any defective parts. --- Make certain the serial lines are not shorted and the common mode voltage does not exceed 7V between any 2 drops.

APPENDIX C: LAN RELATED DRIVE FAULT CODES (continued)

Diagnostic Readout	Fault Code	Name	Description/Causes/Remedies
101100	FLA8	MFRZM	Too Many Parameters Programmed in the Freeze Mask. There are more than 6 parameters programmed in the freeze mask (see EE215, MSET5.) --- Reprogram EE215 to ask for 6 or less parameters.
101101	FLA9	MSMSG	Received Serial Data Insufficient in length. Required number of data bytes (for the received command) were not received from the LAN, but the message checksum was correct. --- Too many sets of termination resistors in the circuit (max = 5, JP18 & 19 tied 1 - 2). --- Serial link lines may have a momentary short. --- Driver of the transmitting device or the receiver of this LAN drop may have failed. --- Serial port or timer of processor (U30) may have failed. --- An invalid message may have been sent to this LAN drop. (Check the message that was sent to the LAN drop displaying the fault for correct data and length.)
101110	FLBO	MSBUF	Not Enough Buffering of Received Bytes. The triple byte buffering scheme is insufficient for the amount of window time [between MUP and shared RAM (8155)]. --- Contact factory.
101111	FLB1	MWIND	The Window Interrupted Itself. The start of the next window is too close to the end of the previous window. --- Contact factory.
110000	FLB2	MSWIN	MUP Did Not Have Shared RAM For Long Enough Time. The window interrupt routine took longer to complete than the drive allotted for. --- Contact factory.
110001	FLB3	MPRGR	Trying to Modify an Illegal Address in MUP. The user tried to modify a protected address in MUP. --- Do not try to modify protected addresses in MUP.
110101	FLB7	LANEA	Weighted Zero Protocol Error. Expected all drops to not use weighted zero checksums and acknowledging of group or broadcast commands but found some did. --- Make all drops compatible. Weighted zero protocol requires REV52 or later MUP in drives and REV13 or later SUP in BIU. --- If adding or updating drops, can configure MUP REV52 to not use weighted zero with ADR211.

APPENDIX D: ASCII CONVERSION TABLE

CHAR.	DEC.	HEX
NUL	0	00
SOH	1	01
STX	2	02
ETX	3	03
EOT	4	04
ENQ	5	05
ACK	6	06
BEL	7	07
BS	8	08
HT	9	09
LF	10	0A
VT	11	0B
FF	12	0C
CR	13	0D
SO	14	0E
SI	15	0F
DLE	16	10
DC1	17	11
DC2	18	12
DC3	19	13
DC4	20	14
NAK	21	15
SYN	22	16
ETB	23	17
CAN	24	18
EM	25	19
SUB	26	1A
ESC	27	1B
FS	28	1C
GS	29	1D
RS	30	1E
US	31	1F
SP	32	20
!	33	21
"	34	22
#	35	23
$	36	24
%	37	25
&	38	26
'	39	27
(	40	28
)	41	29
•	42	2A
+	43	2B
,	44	2C
-	45	2D
.	46	2E
/	47	2F

CHAR.	DEC.	HEX
O	48	30
1	49	31
2	50	32
3	51	33
4	52	34
5	53	35
6	54	36
7	55	37
8	56	38
9	57	39
:	58	3A
;	59	3B
<	60	3C
=	61	3D
>	62	3E
?	63	3F
@	64	40
A	65	41
B	66	42
C	67	43
D	68	44
E	69	45
F	70	46
G	71	47
H	72	48
I	73	49
J	74	4A
K	75	4B
L	76	4C
M	77	4D
N	78	4E
O	79	4F
P	80	50
Q	81	51
R	82	52
S	83	53
T	84	54
U	85	55
V	86	56
W	87	57
X	88	58
Y	89	59
Z	90	5A

CHAR.	DEC.	HEX
[	91	5B
\	92	5C
]	93	5D
^	94	5E
-	95	5F
'	96	60
a	97	61
b	98	62
c	99	63
d	100	64
e	101	65
f	102	66
g	103	67
h	104	68
i	105	69
j	106	6A
k	107	6B
l	108	6C
m	109	6D
n	110	6E
o	111	6F
p	112	70
q	113	71
r	114	72
s	115	73
t	116	74
u	117	75
v	118	76
w	119	77
x	120	78
y	121	79
z	122	7A
{	123	7B
/	124	7C
}	125	7D
~	126	7E
DEL	127	7F

APPENDIX E: GLOSSARY

Application Software
Ladder logic code written for a specific project or application which may include one-shots to trigger LAN Standard functions in addition to other instructions.

BIU Address
Unique number assigned to a drop by which communications sent from the drop to the LAN and directed to the drop from the LAN are identified.

Broadcast Communication
LAN communication which is not addressed to any specific drop or group of drops and is received by all drops on the LAN channel.

Bus Interface Unit (BIU)
The circuit board that mounts in a Series Six PLC I/O rack which provides the hardware and software interface between the programmable logic controller and the Local Area Network.

Central Processing Unit (CPU)
The digital processor in the PLC which interprets and executes instructions.

Channel
A specific LAN link connecting a number of drops with a single processor (SUP) in the BIU. There are two separate channels (P and Q) per BIU card.

Data Processing Request (DPREQ)
Series Six instruction used to access and exchange data with the BIU.

DIP Switch
A group of miniature switches packaged in a group with two parallel rows of connecting pins (Dual In-line Package).

Directed Communication
LAN communication which is addressed to a specific drop.

Drop
A specific device or "member" connected to a Local Area Network channel.

Freeze Command
Command issued by the LAN Standard software to transfer feedback data from the drive.

Group Communication
LAN communication which is addressed to a specific set of drops, each of which has been programmed to respond to that group designation.

Input/Output (I/O)
Connections to the PLC system that interface between user devices (switches, relays, indicators etc.) and the controller.

APPENDIX E: GLOSSARY (continued)

Ladder Logic
Programming used in the Series Six PLC to direct the operation of the controller; so named due to its symbolic equivalence to the ladder diagrams used to represent relay control systems.

LAN Standard Software
Standardized code which serves as the "operating system" for directing LAN communications by the Series Six PLC.

Local Area Network (LAN)
A data communications link connecting several devices together within a predefined physical area through standardized electrical connections and communication protocols.

Microapplication Card
The circuit board that mounts in the card carrier rack of the drive controller which provides the hardware and software interface between the drive and the Local Area Network.

MUP
Microprocessor on the Microapplication card.

One-shot
A ladder logic "coil" which, once energized, will remain on until the completion of the logic scan.

Programmable Logic Controller (PLC)
A digital control system consisting of a digital processor, input/output interfaces and memory which accepts inputs from devices such as switches and senors and provides outputs in response to programmed instructions.

Protocol
A predefined set of rules governing communications procedures and message formatting.

RS-422
An Electronic Industries Association (EIA) standard specification defining the mechanical and electrical characteristics for data communications using differential signals.

RUP
Main regulating microprocessor of the drive.

SUP
Either of two microprocessors contained on the BIU card. There is one SUP per BIU channel, SUP-P and SUP-Q.

Watchdog Timer
An internal timer which is used as a system check.

Window
A period allotted in the PLC processor sweep for data communications or processing.

Zone Communication
LAN communication which is addressed to a series of drops in a predefined zone.



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